

$$m_2 = -\frac{3}{2}$$

$$\tan \theta = \frac{8}{25}$$

Paragraph for Question I02/3/087 to I02/3/088

Let $f(x)$ be a continuous function such that $f(x) = \int_0^x (f(t) \cos t - \cos(t-x)) dt$.

माना $f(x)$ एक सतत् फलन इस प्रकार है कि $f(x) = \int_0^x (f(t) \cos t - \cos(t-x)) dt$.

I02/3/087. Which of the following statement(s) is/are false ?

- (A) $f(x)$ is a periodic function
- (B) Number of roots of equation $f(x) = 0$ in $[0, 2\pi]$ is 3
- (C) Between any two roots of the equation $f(x) = 0$ there exists at least one root of the equation $\cos x = 0$
- (D) $f(x)$ is strictly increasing for $x \in (0, 2\pi)$

निम्न में से कौनसा/से कथन सत्य है/हैं ?

- (A) $f(x)$ एक आवर्ती फलन है
- (B) समीकरण $f(x) = 0$ के अंतराल $[0, 2\pi]$ में हलों की संख्या 3 हैं
- (C) समीकरण $f(x) = 0$ के किन्हीं दो मूलों के मध्य समीकरण $\cos x = 0$ का कम से कम एक मूल विद्यमान है
- (D) $f(x)$, $x \in (0, 2\pi)$ में निरंतर विद्यमान हैं

Ans. D

Sol.

I02/3/088. Number of integers in range of $f(x)$ is :

$f(x)$ के परिसर में पूर्णाकों की संख्या है :

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Ans. B

Sol. $f(x) = \int_0^x f(t) \cos t dt - \int_0^x \cos(t-x) dt$

$$f(x) = \int_0^x f(t) \cos t dt - \int_0^x \cos t dt$$

$$f'(x) = f(x) \cos x - \cos x$$

$$\Rightarrow \frac{f'(x)}{f(x-1)} = \cos x$$

Integrating both sides w.r.t. x , we get

$$\ln|f(x)-1| = \sin x + C$$

$$\text{Now } f(0) = 0$$

$$\Rightarrow c = 0 \Rightarrow |f(x) - 1| = e^{\sin x}$$

$$f(x) = 1 \pm e^{\sin x}$$

$$\text{But } f(0) = 0 \Rightarrow f(x) = 1 - e^{\sin x}$$

Now $f(x)$ is periodic

$$\text{Also } f(x) = 0$$

$$\Rightarrow e^{\sin x} = 1$$

$$\Rightarrow \sin x = 0$$

$$\Rightarrow x = 0, \pi, 2\pi$$

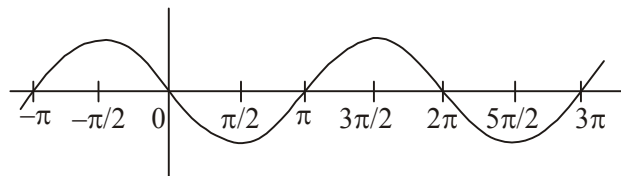
Also suppose $f(x) = 0$ has two roots α & β

$$\therefore f(\alpha) = f(\beta)$$

$\therefore$ Using Rolle's theorem the equation $f'(x) = 0$ has at least one root in $x \in (\alpha, \beta)$

But $f(x)$ is non-monotonic

$\therefore$ (D) is false.



$$\therefore \text{Range of } f(x) \text{ is } \left[1 - e, 1 - \frac{1}{e}\right]$$

$\therefore$ 2 integers.

Paragraph for Question I02/3/089 to I02/3/090

Let $f(x)$ be a polynomial of degree four such that $f(0) = f'(1) = 0$ and $f'''(3) = 24$. Also, $f'' : \mathbb{R} \rightarrow [-4, \infty)$ is a surjective function such that $f''(2-x) = f''(2+x)$.

माना $f(x)$ एक चार घातीय बहुपद इस प्रकार है कि $f(0) = f'(1) = 0$ तथा $f'''(3) = 24$ हैं एवं $f'' : \mathbb{R} \rightarrow [-4, \infty)$ एक surjective फलन इस प्रकार से है कि $f''(2-x) = f''(2+x)$

I02/3/089. The value of $\int_0^1 (f(-x) - f(x)) dx$ is :

$$\int_0^1 (f(-x) - f(x)) dx \text{ का मान है :}$$

(A) 22

(B) 24

(C) 26

(D) 28

Ans. D

Sol.

I02/3/090. The equation of tangent to curve $y = f(x)$ at $x = 2$ is :

वक्र $y = f(x)$ की $x = 2$ पर स्पर्शरेखा का समीकरण होगा :

(A) $y = 0$

(B) $y + 8 = 0$

(C) $x + 8 = 0$

(D) $x + y + 6 = 0$

Ans. B

Sol. $f''(x)$ is symmetric about $x = 2$

$$\therefore f''(x) = a(x - 2)^2 + k$$

$$\therefore f''(x) = 2a(x - 2)$$

$$\because f'''(3) = 24, \therefore a = 12$$

$$\therefore f''(x) = 12(x - 2)^2 - 4 \quad (\because k = -4)$$

$$\therefore f'(x) = 4x^3 - 24x^2 + 44x - 24$$

$$\therefore f(x) = x^4 - 8x^3 + 22x^2 - 24x \quad (f(0) = 0)$$

$$\int_0^1 f(-x) - f(x) dx$$

$$= \int_0^1 16x^3 + 48x dx = 28$$

$$\frac{dy}{dx} \text{ at } x = 2 \text{ is } 0$$

$$\text{At } x = 2 : y = -8$$

$$\therefore \text{Equation : } y + 8 = 0$$

Paragraph for Question I02/3/091 to I02/3/092

Consider the following integrals

$$I = \int_1^2 \frac{1}{x - 1 + \sqrt{x^2 - 3x + 3}} dx \text{ and } J = \int_1^2 \frac{x^2 - x + 1}{x(2x - 1)^2} dx$$

Now, answer the following questions :

$$I = \int_1^2 \frac{1}{x-1+\sqrt{x^2-3x+3}} dx \text{ तथा } J = \int_1^2 \frac{x^2-x+1}{x(2x-1)^2} dx$$

अब, निम्न प्रश्नों के उत्तर दीजिए :

I02/3/091. The value of $I - 2J$ is equal to :

$I - 2J$ का मान बराबर होगा :

- (A) 3 (B) 2 (C) 1 (D) 0

Ans. D

Sol.

I02/3/092. The value of $\text{sgn}(4(I - J) - 1) + 8 + \text{sgn}(2 - 4(I - J))$ is equal to :

$\text{sgn}(4(I - J) - 1) + 8 + \text{sgn}(2 - 4(I - J))$ का मान बराबर होगा :

- (A) 8 (B) 9 (C) 10 (D) 11

Ans. C

Sol.

Paragraph for Question I02/3/093 to I02/3/094

Let

$$\int \frac{x^2}{(x^2-1)\sin 2x + 2x \cos 2x} dx = f(x) + C; f\left(\frac{\pi}{2}\right) = \frac{1}{2} \ln \frac{\pi}{2}$$

Now, answer the following questions :

Here C denotes the constant of integration.

माना

$$\int \frac{x^2}{(x^2-1)\sin 2x + 2x \cos 2x} dx = f(x) + C; f\left(\frac{\pi}{2}\right) = \frac{1}{2} \ln \frac{\pi}{2}$$

अब, निम्न प्रश्नों के उत्तर दीजिए :

यहाँ C समाकलन के अचर को दर्शाता है।

I02/3/093. The integer nearest to value of $e^{-2f(\pi)}$ is equal to :

$e^{-2f(\pi)}$ के मान का निकटतम पूर्णांक होगा :

- (A) 2 (B) 3 (C) 4 (D) 5

Ans. B
Sol.

I02/3/094. The integer nearest to value of $e^{2f\left(\frac{\pi}{4}\right)}$ is equal to :

$e^{2f\left(\frac{\pi}{4}\right)}$ के मान का निकटतम पूर्णांक होगा :

(A) 8 (B) 7 (C) 6 (D) 5

Ans. A
Sol.

Paragraph for Question I03/3/095 to I03/3/096

A differentiable function $y = g(x)$ satisfies $\int_0^x (x-t+1) g(t) dt = x^4 + x^2$; $\forall x \geq 0$

एक अवकलनीय फलन $y = g(x)$ है जो $\int_0^x (x-t+1) g(t) dt = x^4 + x^2$; $\forall x \geq 0$ को संतुष्ट करता है

I03/3/095. $y = g(x)$ satisfies the differential equation :

$y = g(x)$ अवकल समीकरण को संतुष्ट करता है:

(A) $\frac{dy}{dx} - y = 12x^2 + 2$ (B) $\frac{dy}{dx} + 2y = 12x^2 + 2$
(C) $\frac{dy}{dx} + y = 12x^2 + 2$ (D) $\frac{dy}{dx} + y = 12x + 2$

Ans. C

Sol. $x \int_0^x g(t) dt + \int_0^x (1-t) g(t) dt = x^4 + x^2$

differentiate w.r.t. 'x'

$$x g(x) + \int_0^x g(t) dt + (1-x) g(x) = 4x^3 + 2x \quad \dots(1)$$

From (1)

$$\int_0^x g(t) dt + g(x) = 4x^3 + 2x$$

$$\text{let } g(x) + g'(x) = 12x^2 + 2 \Rightarrow \frac{dy}{dx} + y = 12x^2 + 2 \quad (\because y = g(x))$$

I03/3/096. The value of $g(0)$ equals :

- (A) 0 (B) 1 (C) e^2 (D) Data insufficient

$g(0)$ का मान होगा :

- (A) 0 (B) 1 (C) e^2 (D) तथ्य अपर्याप्त

Ans. A

Sol.
$$x \int_0^x g(t) dt + \int_0^x (1-t) g(t) dt = x^4 + x^2$$

differentiate w.r.t. 'x'

$$x g(x) + \int_0^x g(t) dt + (1-x) g(x) = 4x^3 + 2x \quad \dots(1)$$

Put $x = 0$ in (1) we get $g(0) = 0$

Paragraph for Question I03/3/097 to I03/3/099

Suppose f and g are differentiable functions such that $xg(f(x)) f'(g(x))g'(x) = f(g(x)) g'(f(x)) f'(x) \forall x \in \mathbb{R}$ and

f is positive, g is positive $\forall x \in \mathbb{R}$. Also $\int_0^x f(g(t)) dt = \frac{1}{2}(1 - e^{-2x}) \forall x \in \mathbb{R}$, $g(f(0)) = 1$ and $h(x) = \frac{g(f(x))}{f(g(x))} \forall x \in$

$\mathbb{R}$.

I03/3/097. The graph of $y = h(x)$ is symmetric with respect to line :

- (A) $x = -1$ (B) $x = 0$ (C) $x = 1$ (D) $x = 2$

Ans. C

Sol. $f(g(x)) = e^{-2x}$

$$\frac{x \cdot [f(g(x))]' }{f(g(x))} = \frac{[g(f(x))]' }{g[f(x)]}$$

$$\Rightarrow g(f(x)) = e^{-x^2}$$

$$H(x) = e^{-(x-1)^2+1}$$

I03/3/098. The value of $f(g(0)) + g(f(0)) =$:

- (A) 1 (B) 2 (C) 3 (D) 4

Ans. B

Sol. $f(g(0)) + g(f(0)) = 2$

I03/3/099. The largest possible value of $h(x) \forall x \in \mathbb{R}$ is :

- (A) 1 (B) $e^{\frac{1}{3}}$ (C) e (D) e^2

Ans. C

Sol. $H(x)_{\max} = e$

Paragraph for Question I03/3/100 to I03/3/101

Let f be a real-valued differentiable function on $\mathbb{R}$ satisfying $f'(x) + f(x) = x^2 + 3x + 2$ for all $x \in \mathbb{R}$ and $f(q) \in \mathbb{Q}$ for all $q \in \mathbb{Q}$.

[**Note:** $\mathbb{R}$ and $\mathbb{Q}$ denote the set of all real numbers and the set of all rational numbers respectively.]

माना f एक $\mathbb{R}$ में, वास्तविक मान अवकलनीय फलन है, जो समीकरण $f'(x) + f(x) = x^2 + 3x + 2$ सभी $x \in \mathbb{R}$ तथा $f(q) \in \mathbb{Q}$ सभी $q \in \mathbb{Q}$ के लिए संतुष्ट करता है।

[नोट : $\mathbb{R}$ तथा $\mathbb{Q}$ क्रमशः सभी वास्तविक संख्याओं का समुच्चय है तथा सभी प्राकृत संख्याओं का समुच्चय है]

I03/3/100. Which of the following statement is correct :

- (A) $f'(x)$ is strictly decreasing in $[1, \infty)$
(B) $f'(x)$ equals 3 when $x = 2$
(C) There exists $\alpha > 1$ such that $|f(x)| \leq \alpha$ for all $x \in \mathbb{R}$
(D) $f(|x|)$ is non-differentiable at $x = 0$

निम्न में से कौनसा कथन सत्य है :

- (A) $f'(x)$ अन्तराल $[1, \infty)$ में ह्रस्वमान है
(B) जब $x = 2$ है, तो $f'(x) = 3$ है
(C) सभी $x \in \mathbb{R}$ के लिए, $\alpha > 1$ इस प्रकार कि $|f(x)| \leq \alpha$ विद्यमान है
(D) $f(|x|)$, $x = 0$ पर अवकलनीय फलन नहीं है

Ans. D

Sol.

I03/3/101. The area bounded by the curve $y = f(x)$ with x -axis between ordinates $x = 0$ and $x = 1$ equals :

वक्र $y = f(x)$ तथा x -अक्ष एवं निर्देशांक $x = 0$ तथा $x = 1$ द्वारा परिबद्ध क्षेत्र का क्षेत्रफल होगा :

- (A) $\frac{7}{6}$ (B) $\frac{11}{6}$ (C) $\frac{13}{6}$ (D) $\frac{17}{6}$

Ans. B

Sol. Given, $f'(x) + f(x) = x^2 + 3x + 2$

$$\Rightarrow \frac{dy}{dx} + y = x^2 + 3x + 2 \quad (\text{Linear differential equation})$$

Now, $IF = e^{\int 1 dx} = e^x$

$\therefore$ General solution is

$$y \cdot e^x = \int \underbrace{e^x}_{(II)} \underbrace{(x^2 + 3x + 2)}_{\substack{(I) \\ (I.B.P.)}} dx$$

$$\Rightarrow y \cdot e^x = (x^2 + 3x + 2) e^x - \int e^x (2x + 3) dx = e^x (x^2 + 3x + 2 - 2x - 3 + 2) + C$$

$$\Rightarrow y = (x^2 + x + 1) + C e^{-x}$$

As, $f(q) \in Q \forall q \in Q$,

So, $C = 0$

Hence, $y = f(x) = x^2 + x + 1$

Now, verify alternatives.

Aliter : Let $f(x) = ax^2 + bx + c$, as $f(q) \in Q \forall q \in Q$

Put $f(x)$ in equation, $f'(x) + f(x) = x^2 + 3x + 2$

$\therefore$ On comparing, we get $a = 1, b = 1, c = 1$

Hence, $y = f(x) = x^2 + x + 1$.

Now, verify alternatives

Paragraph for Question I01/3/102 to I01/3/103

Let $f : R \rightarrow R$ be such that $f'(0) = 1$ and $f(x + y) = f(x) + f(y) + e^{x+y}(x + y) - xe^x - ye^y + 2xy \forall x, y \in R$.

On the basis of above information, answer the following questions :

माना $f : R \rightarrow R$ इस प्रकार है कि $f'(0) = 1$ तथा $f(x + y) = f(x) + f(y) + e^{x+y}(x + y) - xe^x - ye^y + 2xy \forall x, y \in R$ है।

उपरोक्त जानकारी के आधार पर, निम्न प्रश्नों के उत्तर दीजिए :

I01/3/102. If $\int \frac{f(x)}{x} dx = g(x) + c$, where $g(0) = 0$, then the value of $g(2)$ is equal to :

यदि $\int \frac{f(x)}{x} dx = g(x) + c$, जहाँ $g(0) = 0$ है, तो $g(2)$ का मान होगा :

- (A) $e^2 + 2$ (B) $2e^2$ (C) $e^2 + 1$ (D) $\frac{e^2}{2}$

Ans. C

Sol.

I01/3/103. Number of distinct real roots of the equation $f(x) - 2x^2 = 0$ is equal to :

समीकरण $f(x) - 2x^2 = 0$ के विभिन्न वास्तविक मूलों की संख्या होगी :

- (A) 0 (B) 1 (C) 2 (D) 3

Ans. B

Sol.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(x) + f(h) + e^{x+h}(x+h) - xe^x - he^h + 2xh - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(h)}{h} + \lim_{h \rightarrow 0} \left(xe^x \frac{(e^h - 1)}{h} + e^{x+h} - e^h + 2x \right)$$

$$= f'(0) + xe^x + e^x - 1 + 2x = 1 + xe^x + e^x - 1 + 2x$$

$$\Rightarrow f'(x) = e^x(x+1) + 2x$$

$$\Rightarrow f(x) = x^2 + xe^x + c \quad \because f(0) = 0 \Rightarrow c = 0$$

$$\therefore f(x) = x^2 + xe^x$$

$$\therefore \int \frac{f(x)}{x} dx = \int (x + e^x) dx = \frac{x^2}{2} + e^x + k$$

$$= \left(\frac{x^2}{2} + e^x - 1 \right) + \lambda$$

$$\therefore g(x) = \frac{x^2}{2} + e^x - 1, \text{ clearly } g(0) = 0$$

$$\therefore g(2) = 2 + e^2 - 1 = 1 + e^2$$

$$f(x) - 2x^2 \Rightarrow xe^x = x^2 \text{ (clearly } x = 0 \text{ is a solution)}$$

$$\Rightarrow e^x = x \text{ has no solution.}$$

$$\therefore \text{Number of solutions} = 1$$

Paragraph for Question I01/3/104 to I01/3/106

$$\text{Let } \int (x^2-1)e^x (x^2+4x+1)dx = e^x f(x) + C$$

On the basis of above information, answer the following questions :

$$\text{माना } \int (x^2-1)e^x (x^2+4x+1)dx = e^x f(x) + C$$

उपरोक्त जानकारी के आधार पर, निम्न प्रश्नों के उत्तर दीजिए :

I01/3/104. If α is one irrational root of equation $f(x) + f'(x) = 0$, then $\alpha^4 + 4\alpha^3 + 2\alpha^2 + 4\alpha$ is equal to :

यदि α समीकरण $f(x) + f'(x) = 0$ का एक अपरिमेय मूल है, तो $\alpha^4 + 4\alpha^3 + 2\alpha^2 + 4\alpha$ बराबर होगा :

- (A) 0 (B) 1 (C) -1 (D) 10

Ans. C

Sol. Differentiate both side

$$(x^2 - 1)(x^2 + 4x + 1) = f(x) + f'(x)$$

$$\Rightarrow \alpha^2 + 4\alpha + 1 = 0$$

$$\Rightarrow \alpha^4 + 4\alpha^3 + 2\alpha^2 + 4\alpha = -1$$

I01/3/105. $f(x)$ is :

- (A) strictly increasing in $(-1, \infty)$ (B) strictly decreasing in $(-\infty, 2)$
(C) strictly increasing in $(-\infty, 2)$ (D) strictly decreasing in $(-\infty, 1)$

$f(x)$ होगा :

- (A) $(-1, \infty)$ में निरंतर वर्धमान (B) $(-\infty, 2)$ में निरंतर ह्रसमान
(C) $(-\infty, 2)$ में निरंतर वर्धमान (D) $(-\infty, 1)$ में निरंतर ह्रसमान

Ans. D

Sol. $\int e^x (x^4 - 4x + 3 + 4x^3 - 4) dx = e^x (x^4 - 4x + 3) + C$

$$\Rightarrow f(x) = x^4 - 4x + 3$$

$$f'(x) = 4x^3 - 4$$

$$f'(x) \geq 0 \quad \forall x \in [1, \infty)$$

$$f'(x) \leq 0 \quad \forall x \in (-\infty, 1]$$

I01/3/106. Angle of intersection of curves $y = f(x)$ & $y = f'(x)$ is θ , then $|\tan \theta|$ is equal to :

चक्र $y = f(x)$ तथा $y = f'(x)$ का प्रतिच्छेदन कोण θ है, तो $|\tan \theta|$ बराबर होगा :

- (A) 1 (B) 2 (C) 8 (D) 12

Ans. D

Sol. $f(x) = x^4 - 4x + 3$
 $g(x) = f'(x) = 4x^3 - 4$
 point of intersection is $x = 1$
 $f'(1) = 0$
 $g'(1) = 12$
 $\Rightarrow |\tan \theta| = 12$

Paragraph for Question I01/3/107 to I01/3/108

Repeated application of integration by parts gives us, the reduction formula if the integrand is dependent of n , $n \in \mathbb{N}$.

खण्डशः समाकलन की पुनरावृत्ति प्रयोग से अपचयन सूत्र प्राप्त होते हैं यदि समाकल्य n पर आश्रित है जहाँ $n \in \mathbb{N}$ ।

I01/3/107. If $I_{m-2, n+2} = \int \sin^{m-2} x \cos^{n+2} x dx$ and $I_{m, n} = -\frac{\sin^{m-1} x \cos^{n+1} x}{(n+1)} + f(m, n) I_{m-2, n+2}$, then $f(2, 3)$ is equal :

यदि $I_{m-2, n+2} = \int \sin^{m-2} x \cos^{n+2} x dx$ तथा $I_{m, n} = -\frac{\sin^{m-1} x \cos^{n+1} x}{(n+1)} + f(m, n) I_{m-2, n+2}$, तो $f(2, 3)$ होगा :

- (A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{1}{4}$ (D) $\frac{1}{5}$

Ans. C

Sol. Let $P = -\sin^{m-1} x \cdot \cos^{n+1} x$

$$\therefore \frac{dP}{dx} = \sin^{m-1} x (n+1) \cos^n x (-\sin x) + \cos^{n+1} x \cdot (m-1) \sin^{m-2} x \cdot \cos x$$

$$= -(n+1) \sin^m x \cos^n x + (m-1) \sin^{m-2} x \cos^{n+2} x$$

On integrating both sides, we get

$$(n+1) I_{m, n} = -\sin^{m-1} x \cos^{n+1} x + (m-1) I_{m-2, n+2}$$

$$\text{or } I_{m, n} = -\frac{\sin^{m-1} x \cos^{n+1} x}{(n+1)} + \frac{(m-1)}{(n+1)} I_{m-2, n+2}$$

$$\text{Here } f(m, n) = \frac{(m-1)}{(n+1)} \quad \therefore f(2, 3) = \frac{1}{4}$$

I01/3/108. If $I_{m,p} = \int x^m \cdot (a + bx^n)^p dx$ and $I_{m,p} = \frac{x^{m+1} (a + bx^n)^p}{(m+1)} - f(m, n, p, b) I_{m+n, p-1}$, then the value of $f(1, 2, 3, 4)$

is equal to :

- (A) 8 (B) 10 (C) 12 (D) none of these

यदि $I_{m,p} = \int x^m \cdot (a + bx^n)^p dx$ तथा $I_{m,p} = \frac{x^{m+1} (a + bx^n)^p}{(m+1)} - f(m, n, p, b) I_{m+n, p-1}$, तब $f(1, 2, 3, 4)$ का मान

बराबर होगा :

- (A) 8 (B) 10 (C) 12 (D) इनमें से कोई नहीं

Ans. C

Sol. Let $P = x^{m+1} (a + bx^n)^p$

$$\therefore \frac{dP}{dx} = x^{m+1} \cdot p(a + bx^n)^{p-1} \cdot (bnx^{n-1}) + (a + bx^n)^p \cdot (m+1)x^m$$

$= bpnx^{m+n} \cdot (a + bx^n)^{p-1} + (m+1)x^m (a + bx^n)^p$ on, integrating both sides, we get

$$P = bpnI_{m+n, p-1} + (m+1)I_{m,p} \text{ or } I_{m,p} = \frac{x^{m+1} (a + bx^n)^p}{(m+1)} - \frac{bnp}{(m+1)}$$

$$\Rightarrow f(1, 2, 3, 4) = \frac{2 \times 3 \times 4}{1+1} = 12$$

Paragraph for Question I01/3/109 to I01/3/111

Let f and g be two differentiable functions such that

$$f(x) = g'(1) \sin x + (g''(2) - 1)x$$

$$g(x) = x^2 - f'\left(\frac{\pi}{2}\right)x + f''\left(-\frac{\pi}{2}\right)$$

माना f तथा g दो अवकलनीय फलन इस प्रकार हैं कि

$$f(x) = g'(1) \sin x + (g''(2) - 1)x$$

$$g(x) = x^2 - f'\left(\frac{\pi}{2}\right)x + f''\left(-\frac{\pi}{2}\right)$$

I01/3/109. The number of solution(s) of the equation $f(x) = g(x)$ is/are :

- (A) 1 (B) 2 (C) 3 (D) infinite

समीकरण $f(x) = g(x)$ के हलों की संख्या होगी :

(A) 1

(B) 2

(C) 3

(D) अनन्त

Ans. B**Sol.**

I01/3/110. If $\int \frac{g(\cos x)}{f(x) - x} dx = \cos x + \ln(h(x)) + C$ where C is constant and $h\left(\frac{\pi}{2}\right) = 1$ then $\left| h\left(\frac{2\pi}{3}\right) \right|$ is :

यदि $\int \frac{g(\cos x)}{f(x) - x} dx = \cos x + \ln(h(x)) + C$ जहाँ C नियतांक है तथा $h\left(\frac{\pi}{2}\right) = 1$ है तब $\left| h\left(\frac{2\pi}{3}\right) \right|$ होगा :

(A) $3\sqrt{2}$ (B) $2\sqrt{3}$ (C) $\sqrt{3}$ (D) $\frac{1}{\sqrt{3}}$ **Ans.** B**Sol.**

I01/3/111. If $\phi(x) = f^{-1}(x)$ then $\phi'\left(\frac{\pi}{2} + 1\right)$ equals to :

यदि $\phi(x) = f^{-1}(x)$ तब $\phi'\left(\frac{\pi}{2} + 1\right)$ बराबर होगा :

(A) $\frac{\pi}{2} + 1$ (B) $\frac{\pi}{2}$

(C) 1

(D) 0

Ans. C**Sol.** $f(x) = g'(1)\sin x + (g''(2) - 1)x$

$$\Rightarrow f'(x) = g'(1)\cos x + g''(2) - 1 \Rightarrow f'\left(\frac{\pi}{2}\right) = g''(2) - 1$$

$$f''(x) = -g'(1)\sin x \Rightarrow f''\left(\frac{\pi}{2}\right) = -g'(1)$$

$$g(x) = x^2 - f'\left(\frac{\pi}{2}\right)x + f''\left(-\frac{\pi}{2}\right) \Rightarrow g'(x) = 2x - f'\left(\frac{\pi}{2}\right)$$

$$g''(x) = 2 \Rightarrow g''(2) = 2$$

$$f(x) = \sin x + x \quad \text{and} \quad g(x) = x^2 - x + 1$$

INTEGER TYPE QUESTIONS

I04/4/112 The set of points (x, y) in the plane satisfying $x^{2/5} + |y| = 1$ form a curve enclosing a region of area $\frac{p}{q}$ square units, where p and q are relatively prime positive integers. Find $p + q$.

समतल में बिन्दुओं (x, y) का समुच्चय जो $x^{2/5} + |y| = 1$ को सन्तुष्ट करते हैं एक वक्र का निर्माण करते हैं जिसका क्षेत्रफल $\frac{p}{q}$ वर्ग इकाई है जहाँ p तथा q सापेक्षिक अभाज्य धनात्मक पूर्णांक हैं तो $p + q$ का मान ज्ञात किजिये ?

Ans. 15

Sol.
$$Ar = 4 \int_0^1 (1 - x^{2/5}) dx = 4 \left(x - \frac{5}{7} x^{7/5} \right)_0^1 = \frac{8}{7}$$

I04/4/113. Let $f: [0, 2\pi] \rightarrow \mathbb{R}$ be given as $f(x) = \sin x \forall x \in \left[0, \frac{\pi}{2}\right]$, $f(x) + f(\pi - x) = 2 \forall x \in \left(\frac{\pi}{2}, \pi\right]$ and $f(x) = f(2\pi - x) \forall x \in (\pi, 2\pi)$, then area bounded by curve $y = f(x)$ and x -axis is $k\pi$ square units where k is :

माना $f: [0, 2\pi] \rightarrow \mathbb{R}$, $f(x) = \sin x \forall x \in \left[0, \frac{\pi}{2}\right]$, $f(x) + f(\pi - x) = 2 \forall x \in \left(\frac{\pi}{2}, \pi\right]$ तथा $f(x) = f(2\pi - x) \forall x \in (\pi, 2\pi)$ द्वारा दिया गया है, तो वक्र $y = f(x)$ तथा x -अक्ष द्वारा परिबद्ध क्षेत्रफल $k\pi$ वर्ग इकाई है, जहाँ k का मान होगा :

Ans. 2

Sol.
$$f(x) = \begin{cases} \sin x & x \in \left[0, \frac{\pi}{2}\right] \\ 2 - \sin x & x \in \left(\frac{\pi}{2}, \pi\right] \\ 2 + \sin x & x \in \left(\pi, \frac{3\pi}{2}\right] \\ -\sin x & x \in \left(\frac{3\pi}{2}, 2\pi\right] \end{cases}$$

$$\text{Area} = \int_0^{\frac{\pi}{2}} \sin x dx + \int_{\frac{\pi}{2}}^{\pi} 2 - \sin x dx$$

$$+ \int_{\pi}^{\frac{3\pi}{2}} 2 + \sin x dx + \int_{\frac{3\pi}{2}}^{2\pi} -\sin x dx$$

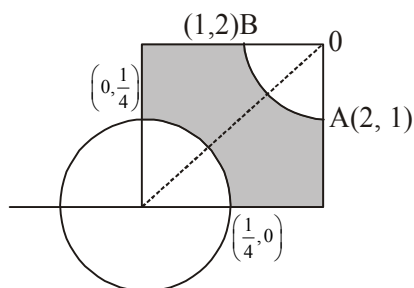
$$= 1 + \pi - 1 + \pi - 1 + 1 = 2\pi \text{ sq. units.}$$

I04/4/114. If the area bounded by x-axis, y-axis, $x = 2$, $y = 2$ & $xy \leq 2 \leq 8(x^2 + y^2)$ is $\alpha - \frac{\pi}{\beta} + \ell n \gamma$ where $\alpha, \beta, \gamma \in \mathbb{N}$ then $\beta - 2(\alpha + \gamma)$ is equal to _____.

यदि x-अक्ष, y-अक्ष, $x = 2$, $y = 2$ तथा $xy \leq 2 \leq 8(x^2 + y^2)$ से परिबद्ध क्षेत्रफल $\alpha - \frac{\pi}{\beta} + \ell n \gamma$ है जहाँ $\alpha, \beta, \gamma \in \mathbb{N}$ है तब $\beta - 2(\alpha + \gamma)$ बराबर होगा _____.

Ans. 4

Sol.



$xy = 2$ is rectangular hyperbola & $(x^2 + y^2) \geq 2$ is exterior of circle; shaded portion is given by figure.

$$\text{Area of portion OAB} = 2 - \int_1^2 \frac{2}{x} dx = (2 - \ell n 4)$$

$$\text{Total area} = 4 - \left(\frac{\pi}{16} + 2 - \ell n 4 \right)$$

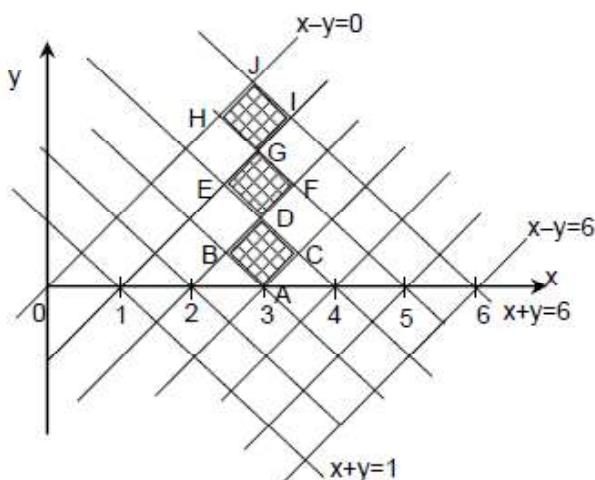
$$2 + \ell n 4 - \frac{\pi}{16}$$

I04/4/115. If A is the area enclosed by the curve $[x + y] + [x - y] = 5$, for $x \geq y \forall x, y \geq 0$, then 4A is _____.

यदि $x \geq y \forall x, y \geq 0$ के लिए वक्र $[x + y] + [x - y] = 5$ द्वारा घिरा हुआ क्षेत्रफल A है, तब 4A होगा _____.

Ans. 6

Sol. $[x + y] + [x - y] = 5$
for $[x - y] = 0$, $[x + y] = 5$
 $\Rightarrow 0 \leq x - y < 1$, $5 \leq x + y < 6$



similarly for $1 \leq x - y < 2$, $4 \leq x + y < 5$ and so on

Required area is shaded region which is area of squares (ABDC + DEGF + GHJI)

$$= 3 \left(\frac{1}{2} \cdot 1 \cdot 1 \right) = \frac{3}{2} \text{ square units}$$

I04/4/116. The area of the region enclosed by the curve $5x^2 + 6xy + 2y^2 + 7x + 6y + 6 = 0$ is equal to the area of the circle $4x^2 + 4y^2 = r^2$ then r^2 is equal to _____.

वक्र $5x^2 + 6xy + 2y^2 + 7x + 6y + 6 = 0$ से घिरे क्षेत्र का क्षेत्रफल वृत्त $4x^2 + 4y^2 = r^2$ के क्षेत्रफल के बराबर है तब r^2 का मान होगा _____.

Ans. 2

Sol. $5x^2 + 6xy + 2y^2 + 7x + 6y + 6 = 0$ is equation of ellipse

$$\text{Now, } y = \frac{-3(1+x) \pm \sqrt{(x-1)(3-x)}}{2}$$

Now for real value of y ; $x \in [1, 3]$

$$\text{Thus, the required area } A = \left| \int_1^3 (y_1 - y_2) dx \right|$$

$$A = \left| -\int_1^3 \sqrt{(3-x)(x-1)} dx \right| \Rightarrow A = \int_1^3 \sqrt{1-(x-2)^2} dx = \frac{\pi}{2} \text{ sq. unit}$$

$$\text{Now, area of circle; } (4x^2 + 4y^2 = r^2) \text{ is } \frac{\pi r^2}{4} \Rightarrow r^2 = 2$$

I04/4/117. A curve $y = f(x)$ is such that $f(x) \geq 0$ and $f(0) = 0$ and bonds a curve linear trapezoid with the base $[0, x]$

whose area is proportional to $(n + 1)^{\text{th}}$ power of $f(x)$. If $f(1) = 1$, then $\{f(10)\}^n$ is k , then the value of $\frac{k}{4}$ is _____.

एक वक्र $y = f(x)$ इस प्रकार है कि $f(x) \geq 0$ तथा $f(0) = 0$ और एक वक्ररेखीय समलम्ब चतुर्भुज को आधार $[0, x]$ से जोड़ता है जिसका क्षेत्रफल $f(x)$ के $(n + 1)$ वें घात के समानुपाती है। यदि $f(1) = 1$, तब $\{f(10)\}^n$ का मान k है। तो $\frac{k}{4}$ का मान होगा _____.

Ans. 2.5

Sol. Area of curve linear trapezoid $OABC = \int_0^x f(x) dx$

$$\text{According to } \int_0^x f(x) dx = k \{f(x)\}^{n+1}$$

Where k is constant of proportionality

Differentiating both sides w.r.t. x

$$f(x) = k(n + 1) [f(x)]^n f'(x)$$

$$\text{Or } \{f(x)^{n-1}\} f'(x) = \frac{1}{k(n+1)}$$

Integrating both sides w.r.t. x

$$\frac{\{f(x)\}^n}{n} = \frac{x}{k(n+1)} + c$$

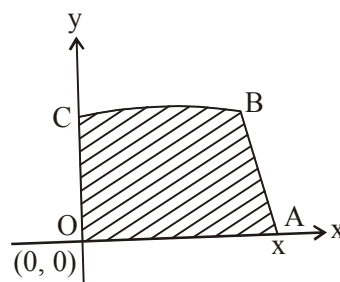
$$\text{Putting } x = 0, \frac{\{f(0)\}^n}{n} = 0 + c$$

$$\Rightarrow 0 = 0 + c \quad [\because f(0) = 0]$$

$$\therefore \frac{\{f(x)\}^n}{n} = \frac{x}{k(n+1)} \Rightarrow \{f(x)\}^n = \frac{nx}{k(n+1)}$$

$$\text{Again, putting } x = 1, \text{ then } \{f(1)\}^n = \frac{n}{k(n+1)} = 0$$

$$\text{From equation (1) } \{f(x)\}^n = x, \{f(10)\}^n = 10$$



I04/4/118. Find the area of the region in the XY plane consisting of all points in the set $\{(x, y) | x^2 + y^2 \leq 144 \text{ and } \sin(2x + 3y) \leq 0\}$ _____.

XY समतल में समुच्चय $\{(x, y) | x^2 + y^2 \leq 144 \text{ तथा } \sin(2x + 3y) \leq 0\}$ के सभी बिन्दुओं को निहित करने वाले क्षेत्र का

क्षेत्रफल होगा

Ans. 72π

Sol. The area of the circular region $S = \{(x, y) | x^2 + y^2 \leq 144\}$ is 144π . The condition $\sin(2x + 3y) \geq 0$ is equivalent to $2x + 3y$ being in one of the intervals $[k\pi, (k+1)\pi]$, where k is an odd integer. The key point is that due to the symmetry of the circle S about any diameter, in particular the diameter $2x + 3y = 0$, the strip inside S lying between the lines $2x + 3y = k\pi$ and $2x + 3y = (k+1)\pi$ is the mirror image of strip lying between the lines $2x + 3y = -k\pi$ and $2x + 3y = -(k+1)\pi$. For each integer k , precisely one of these two equal strips is included in the desired area. Thus the desired area is half that of S , i.e., 72π .

I04/4/119. A spider starts at the origin and runs in the first quadrant along the graph of $y = x^3$ at the constant speed of 10 unit/second. The speed is measured along the length of the curve $y = x^3$. The formula for the curve length along the graph of $y = f(x)$ from $x = a$ to $x = b$ is $\ell = \int_a^b \sqrt{1 + f'(x)^2} dx$. As the spider runs, it spins out a thread that is always maintained in a straight line connecting the spider with the origin. What is the rate in unit/second at which the thread is elongating when the spider is at $\left(\frac{1}{2}, \frac{1}{8}\right)$? _____.

एक मकड़ी मूल बिन्दु से चलना शुरू करती है और पहले चतुर्थांश में $y = x^3$ के ग्राफ के साथ 10 इकाई/सैकण्ड की निरन्तर गति से चलती रहती है। गति को वक्र $y = x^3$ की लम्बाई के अनुदिश मापा जाता है। $x = a$ से $x = b$ तक $y = f(x)$ के रेखाचित्र के अनुदिश वक्र की लम्बाई का सूत्र $\ell = \int_a^b \sqrt{1 + f'(x)^2} dx$ है। जैसे ही मकड़ी दौड़ती है, वह एक धागा बुनती है जो मकड़ी को

हमेशा मूल बिन्दु से जोड़ने वाली एक सीधी रेखा में बना रहता है। जब मकड़ी $\left(\frac{1}{2}, \frac{1}{8}\right)$ पर हो तो धागे के लम्बा होने की दर इकाई/सैकण्ड में क्या होगी _____.

Ans. $\frac{38}{\sqrt{17}}$

Sol. You should use the following names for variables. At any given time t , the spider is at the point (u, u^3) , the length of the thread joining it to the origin in a straight line is s and the curve length along $y = x^3$ from the origin till (u, u^3) is ℓ . You are asked to find $\frac{ds}{dt}$ when $u = \frac{1}{2}$. (Do not try to evaluate the integral for ℓ : it is unnecessary and any attempt to do so will not get any credit because a closed formula in terms of basic functions does not exist.)

$s^2 = u^2 + u^6$. Differentiate with respect to t to get $2s \frac{ds}{dt} = (2u + 6u^5) \frac{du}{dt}$. Plug in $u = \frac{1}{2}$ to get $s = \frac{\sqrt{17}}{8}$

and $\frac{ds}{dt} = \frac{19}{4\sqrt{17}} \frac{du}{dt}$ at the point of interest. To calculate $\frac{du}{dt}$, we use ℓ . By chain rule $\frac{d\ell}{du} = \frac{d\ell}{du} \frac{du}{dt}$. We

are given $\frac{d\ell}{du} = 10$. By the fundamental theorem of calculus, $\frac{d\ell}{du} = \sqrt{1 + 9u^4} = \frac{5}{4}$ at $u = \frac{1}{2}$. Using these

values first get $\frac{du}{dt} = 8$ and then $\frac{ds}{dt} = \frac{38}{\sqrt{17}}$ unit/second.

I04/4/120. Let A_n represents the area formed by the elements of set A, defined below.

$$A = \{(x, y) \in \mathbb{R} \times \mathbb{R} : y \leq [x] + \sqrt{\{x\}}, y \geq 0, 0 \leq x \leq n, n \in \mathbb{N}\}$$

If $[.]$ denotes the greatest integer function and $\{.\}$ denotes the fractional part function, then find the value of $A_3 + A_6$.

माना A_n समुच्चय A के अवयवों से बनने वाले क्षेत्रफल को परिभाषित करता है जहाँ

$$A = \{(x, y) \in \mathbb{R} \times \mathbb{R} : y \leq [x] + \sqrt{\{x\}}, y \geq 0, 0 \leq x \leq n, n \in \mathbb{N}\}$$

यदि $[.]$ महत्तम पूर्णांक फलन तथा $\{.\}$ भिन्नात्मक भाग फलन को दर्शाता है, तब $A_3 + A_6$ का मान ज्ञात कीजिए।

Ans. 24

Sol. $y \leq [x] + \sqrt{\{x\}}, y \geq 0, 0 \leq x \leq n, n \in \mathbb{N}$

$$A_n = \int_0^1 \sqrt{x} dx + \int_1^2 (1 + \sqrt{x-1}) dx + \int_2^3 (2 + \sqrt{x-2}) dx + \dots + \int_{n-1}^n ((n-1) + \sqrt{x-(n-1)}) dx$$

$$A_n = \left[\frac{2}{3} (x)^{\frac{3}{2}} \right]_0^1 + \left[x + \frac{2}{3} (x-1)^{\frac{3}{2}} \right]_1^2 + \dots +$$

$$A_n = \frac{2}{3} + \left[(2-1) + \frac{2}{3} (2-1)^{\frac{3}{2}} \right] + \left[2(3-2) + \frac{2}{3} (3-2)^{\frac{3}{2}} \right] +$$

$$\dots + (n-1) \left[n - (n-1) \right] + \frac{2}{3} \left(n - (n-1) \right)^{\frac{3}{2}}$$

$$A_n = \frac{2}{3} + \left(1 + \frac{2}{3} \right) + \left(2 + \frac{2}{3} \right) + \dots + \left[(n-1) + \frac{2}{3} \right]$$

$$A_n = \frac{2}{3} (n) + \frac{(n-1)(n)}{2}$$

I04/4/121. The functions $f(x)$ and $g(x)$ are defined as

$$f(x) = \begin{cases} -2, & -3 \leq x \leq 0 \\ x-2, & 0 < x \leq 3 \end{cases}, \text{ and}$$

$$g(x) = \min(f(|x|) + |f(x)|, f(|x|) - |f(x)|).$$

The area bounded by $y = g(x)$, $y = 0$, $x = 3$, $x = -3$, is equal to

फलन $f(x)$ तथा $g(x)$ इस प्रकार परिभाषित हैं कि

$$f(x) = \begin{cases} -2, & -3 \leq x \leq 0 \\ x-2, & 0 < x \leq 3 \end{cases}, \text{ तथा}$$

$$g(x) = \min(f(|x|) + |f(x)|, f(|x|) - |f(x)|).$$

$y = g(x)$, $y = 0$, $x = 3$, $x = -3$ से परिबद्ध क्षेत्रफल का मान होगा

Ans. 11.5

Sol.
$$g(x) = \begin{cases} -x-4; & -3 \leq x \leq 0 \\ 2x-4; & 0 < x \leq 2 \\ 0; & 2 < x \leq 3 \end{cases}$$

I02/4/122. If the two lines $AB : \left(\int_0^{2t} \left(\frac{\sin x}{x} + 1 \right) dx \right) x + y = 3t$ and $AC : 2tx + y = 0$ intersect at a point A, then

x-coordinate of point A as $t \rightarrow 0$, is equal to $\frac{p}{q}$ (p and q are in their lowest form). Find $(p+q)$. _____.

यदि दो रेखाएँ $AB : \left(\int_0^{2t} \left(\frac{\sin x}{x} + 1 \right) dx \right) x + y = 3t$ तथा $AC : 2tx + y = 0$ बिन्दु A पर प्रतिच्छेद करती हों, यदि $t \rightarrow$

0 पर बिन्दु A का x-निर्देशांक $\frac{p}{q}$ है (यदि p तथा q अपने सरलतम रूप में हों)। तब $(p+q)$ का मान ज्ञात कीजिए।

Ans. 3/2

Sol.
$$x_A = \lim_{t \rightarrow 0} \frac{3t}{\int_0^{2t} \left(\frac{\sin x}{x} + 1 \right) dx - \int_0^{2t} 1 \cdot dx} = 3; \quad \lim_{t \rightarrow 0} \frac{t}{\int_0^{2t} \frac{\sin x}{x} dx} = \frac{3}{2 \frac{\sin 2t}{2t}} = \frac{3}{2}$$

I02/4/123. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function such that for any $x, y \in [0, 1]$

$$x f(y) + y f(x) \leq 1$$

The maximum value of $\int_0^1 \frac{f(x)}{\pi} dx$ can be expressed as $\frac{p}{q}$ where p and q are relatively prime positive

integers. Find the value of $(p + q)$. _____.

माना $f: [0, 1] \rightarrow \mathbb{R}$ एक सतत् फलन इस प्रकार है कि किसी भी $x, y \in [0, 1]$ के लिए

$$x f(y) + y f(x) \leq 1$$

$\int_0^1 \frac{f(x)}{\pi} dx$ का अधिकतम मान $\frac{p}{q}$ जहाँ p तथा q परस्पर अभाज्य धनात्मक पूर्णांक हैं। $(p + q)$ का मान ज्ञात कीजिए।

Ans.

$$\int_0^{\pi/2} f(\cos t) \cdot \sin t \cdot dt$$

Sol.

Let $x = \sin t$

$$x \in [0, 1] \Rightarrow t \in \left[0, \frac{\pi}{2}\right]$$

$$I = \int_0^1 f(x) dx = \int_0^{\pi/2} f(\sin t) \cdot \cos t \cdot dt = \int_0^{\pi/2} f(\cos t) \cdot \sin t \cdot dt$$

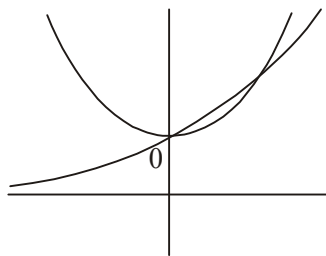
I02/4/124. If $f(x) = 1 + 3 \int_0^x t^2(t) dt$, then the number of solution of $f(x) = x^2 + 1$ is _____.

यदि $f(x) = 1 + 3 \int_0^x t^2(t) dt$ हो, तो $f(x) = x^2 + 1$ के हलों की संख्या होगी।

Ans.

2

Sol.



$$f(x) = 1 + 3 \int_0^x t^2 f(t) dt$$

$$f'(x) = 3(x^2 f(x))$$

$$\Rightarrow \frac{dy}{dx} = 3x^2 dx$$

$$\ln y = x^3 \Rightarrow y = e^{x^3}$$

acc to ques $e^{x^3} = x^2 + 1$

has 2 solutions.

I02/4/125. Let f be a differentiable function satisfying $f'(x) = f^2(x)$ and $f(0) = -1$.

If $\frac{\sqrt{2}}{\pi} \int_0^1 \left(\frac{(1-x)}{f(x)(x^4+1)} + \frac{2}{x^4+1} \right) dx = \frac{p}{q}$ where $p, q \in \mathbb{N}$ then find least value of $(p+q)$ _____.

माना f एक अवकलनीय फलन $f'(x) = f^2(x)$ तथा $f(0) = -1$ को संतुष्ट करता है।

यदि $\frac{\sqrt{2}}{\pi} \int_0^1 \left(\frac{(1-x)}{f(x)(x^4+1)} + \frac{2}{x^4+1} \right) dx = \frac{p}{q}$ जहाँ $p, q \in \mathbb{N}$ तो $(p+q)$ का न्यूनतम मान ज्ञात करें।

Ans. 3

Sol. $f'(x) = f^2(x) \Rightarrow \frac{f'(x)}{f^2(x)} = 1 \Rightarrow \int \frac{f'(x)}{f^2(x)} dx$

$$\int 1 \cdot dx + C \Rightarrow -\frac{1}{f(x)} = x + C$$

$$\text{at } x = 0, f(0) = -1 \Rightarrow C = 1$$

$$\therefore f(x) = -\frac{1}{x+1}$$

$$I = \frac{\sqrt{2}}{\pi} \int_0^1 \left(\frac{1-x}{-\frac{1}{x+1}(x^4+1)} + \frac{2}{x^4+1} \right) dx$$

$$= \frac{\sqrt{2}}{\pi} \int_0^1 \left(\frac{x^2-1}{x^4+1} + \frac{2}{x^4+1} \right) dx = \frac{\sqrt{2}}{\pi} \int_0^1 \frac{x^2+1}{x^4+1} dx$$

$$= \frac{\sqrt{2}}{\pi} \int_0^1 \frac{1 + \frac{1}{x^2}}{x^2 + \frac{1}{x^2}} dx = \frac{\sqrt{2}}{\pi} \int_0^1 \frac{\left(1 + \frac{1}{x^2}\right) dx}{\left(x - \frac{1}{x}\right)^2 + 2}$$

$$x - \frac{1}{x} = t \Rightarrow \left(1 + \frac{1}{x^2}\right) dx = dt$$

$$= \frac{\sqrt{2}}{\pi} \int_{-\infty}^0 \frac{dt}{t^2+2} = \frac{\sqrt{2}}{\pi} \frac{1}{\sqrt{2}} \left(\tan^{-1} \left(\frac{t}{\sqrt{2}} \right) \right)_{-\infty}^0$$

$$= \frac{1}{\pi} \left(0 - \left(-\frac{\pi}{2} \right) \right) = \frac{1}{2} = \frac{p}{q} \quad \therefore (p+q) = 3$$

I02/4/126. If $f(x)$ is a differentiable function satisfying $\int_1^x f(t)dt + \int_2^{f(x)} f^{-1}(t)dt = x^2 f'(x) - 2, x \neq 0, f(1) = 2$, then $f'(2)$ is _____.

यदि $f(x)$ एक अवकलनीय फलन $\int_1^x f(t)dt + \int_2^{f(x)} f^{-1}(t)dt = x^2 f'(x) - 2, x \neq 0, f(1) = 2$ को संतुष्ट करता है, तो $f'(2)$

का मान होगा।

Ans. $\ln 2$

Sol. $xf(x) - 2 = x^2 f'(x) - 2$

$$\left(\because \int_a^b f(x)dx + \int_{f(a)}^{f(b)} f^{-1}(x)dx = bf(b) - af(a) \right)$$

$$\Rightarrow xf'(x) = f(x) \quad (\because x \neq 0)$$

$$\Rightarrow f(x) = 2x$$

$$\Rightarrow \frac{f'(x)}{f(x)} = \frac{1}{x} (\because f(x) \neq 0)$$

$$\Rightarrow \log f(x) = \ln x + C$$

$$\because f(1) = 2 \Rightarrow c = \ln 2$$

I02/4/0127. Let 'n' be a non-negative integer and $I(n) = \int_0^\pi \frac{2 + 2\cos x - \cos(n-1)x - 2\cos nx - \cos(n+1)x}{1 - \cos 2x} dx$. Find the unit

digit in the real number $\left(\frac{I(7)}{\pi} \right)$. _____.

माना कि 'n' एक अऋणात्मक पूर्णांक है तथा $I(n) = \int_0^\pi \frac{2 + 2\cos x - \cos(n-1)x - 2\cos nx - \cos(n+1)x}{1 - \cos 2x} dx$ है। तब वास्तविक

संख्या $\left(\frac{I(7)}{\pi} \right)$ में इकाई का अंक ज्ञात कीजिए।

Ans. $n\pi$

Sol. On simplifying we get

$$I(n) = \int_0^{\pi} \frac{1 - \cos nx}{1 - \cos x} dx = \int_0^{\pi} \frac{\sin^2 \frac{nx}{2}}{\sin^2 \frac{x}{2}} dx$$

$$I(0) = 0$$

$$I(1) = \pi$$

$$I(2) = 2\pi$$

$$I(n) - I(n-1) = \int_0^{\pi} \frac{\sin(2n-1) \cdot \frac{x}{2}}{\sin \frac{x}{2}} dx$$

$$I(n+1) - I(n) = \int_0^{\pi} \frac{\sin(2n+1) \cdot \frac{x}{2}}{\sin \frac{x}{2}} dx$$

$$I(n+1) - 2I(n) + I(n-1) = 2 \int_0^{\pi} \cos nx \cdot dx = 2 \left[\frac{\sin nx}{n} \right]_0^{\pi} = 0$$

$$\Rightarrow I(n-1), I(n), I(n+1) \text{ are in A.P. } \quad (n \in \mathbb{N})$$

$$\therefore I(n) = n\pi$$

I02/4/128. Let $I_n = \int_0^{\pi} \frac{\sin\left(n + \frac{1}{2}\right)x}{\sin\left(\frac{x}{2}\right)} dx$ where $n \in \mathbb{W}$. If $I_1^2 + I_2^2 + I_3^2 + \dots + I_{20}^2 = m\pi^2$ then find the largest prime factor of m . _____.

माना कि $I_n = \int_0^{\pi} \frac{\sin\left(n + \frac{1}{2}\right)x}{\sin\left(\frac{x}{2}\right)} dx$ जहाँ $n \in \mathbb{W}$ है। यदि $I_1^2 + I_2^2 + I_3^2 + \dots + I_{20}^2 = m\pi^2$ है तब m का महत्तम

अभाज्य गुणनखण्ड होगा।

Ans. 5

Sol.
$$I_{n+1} - I_n = \int_0^{\pi} \frac{\sin\left(n + \frac{3}{2}\right)x - \sin\left(n + \frac{1}{2}\right)x}{\sin\left(\frac{x}{2}\right)} dx = \int_0^{\pi} \frac{2 \sin \frac{x}{2} \cdot \cos(n+1)x}{\sin \frac{x}{2}} dx$$

$$= \left[\frac{2 \sin(n+1)x}{n+1} \right]_0^\pi = 0 \quad \therefore I_{n+1} = I_n \quad \therefore I_1 = I_2 = I_3 = \dots = I_n = \text{constant} = \int_0^\pi \frac{\sin \frac{x}{2}}{\sin \frac{x}{2}} dx = \pi$$

$$\therefore I_1^2 + I_2^2 + I_3^2 + \dots + I_{20}^2 = \pi^2 + \pi^2 + \pi^2 + \dots + \pi^2 = 20\pi^2 = m\pi \quad \therefore \text{largest prime factor of } m = 5$$

I02/4/129. Let $I_n = \int_0^{\pi/4} \tan^n x \, dx$; $n = 0, 1, 2, \dots$ and $S_n = \sum_{n=0}^n (I_n I_{n+1} + I_n I_{n+3} + I_{n+1} I_{n+2} + I_{n+2} I_{n+3})$. Find

the value of $\lim_{n \rightarrow \infty} S_n$. _____.

$$\text{माना } I_n = \int_0^{\pi/4} \tan^n x \, dx; n = 0, 1, 2, \dots \text{ तथा } S_n = \sum_{n=0}^n (I_n I_{n+1} + I_n I_{n+3} + I_{n+1} I_{n+2} + I_{n+2} I_{n+3}) \text{ है। } \lim_{n \rightarrow \infty} S_n$$

का मान ज्ञात कीजिए।

Ans. 1

Sol.
$$I_n + I_{n+2} = \int_0^{\pi/4} \tan^n x + \int_0^{\pi/4} \tan^{n+2} x \, dx = \int_0^{\pi/4} \tan^n x \cdot \sec^2 x \, dx = \frac{1}{n+1}$$

$$S_n = \sum_{n=0}^n (I_n + I_{n+2})(I_{n+1} + I_{n+3})$$

$$S_n = \sum_{n=0}^n \frac{1}{(n+1)(n+2)} = \sum_{n=0}^n \left(\frac{1}{n+1} - \frac{1}{n+2} \right)$$

$$S_n = 1 - \frac{1}{n+2}$$

$$\therefore \lim_{n \rightarrow \infty} S_n = 1$$

I02/4/130. For positive t , let α, β be the root of the quadratic equation $x^2 + t^2x - 2t = 0$. If minimum value of

$\int_{-1}^2 \left(\left(x + \frac{1}{\alpha^2} \right) \left(x + \frac{1}{\beta^2} + \frac{1}{\alpha\beta} \right) \right) dx$ is $\sqrt{p} + q$ where p is a rational number and $q \in \mathbb{N}$, then the value of $[p] + q$ is (where $[.]$ denotes G.I.F.) _____.

धनात्मक t के लिए, माना α, β समीकरण $x^2 + t^2x - 2t = 0$ के मूल हैं। यदि $\int_{-1}^2 \left(\left(x + \frac{1}{\alpha^2} \right) \left(x + \frac{1}{\beta^2} + \frac{1}{\alpha\beta} \right) \right) dx$ का

न्यूनतम मान $\sqrt{p} + q$ द्वारा दिया जाता है जहाँ p एक परिमेय संख्या है तथा $q \in \mathbb{N}$ तो $[p] + q$ का मान ज्ञात कीजिए (जहाँ $[.]$ महत्त्वपूर्णक को प्रदर्शित करता है।)

Ans. 4

Sol.
$$\left(x + \frac{1}{\alpha^2}\right)\left(x + \frac{1}{\beta^2}\right) + \frac{1}{\alpha\beta} - x^2 + \left(\frac{\alpha^2 + \beta^2}{(\alpha\beta)^2}\right)x + \frac{1}{(\alpha\beta)^2} + \frac{1}{\alpha\beta} = x^2 + \left(\frac{t^4 + 4t}{4t^2}\right)x + \frac{1}{4t^2} + \frac{1}{2t}$$

Let $I = \int_{-1}^2 x^2 dx + \frac{3}{2} \left(\frac{t^2}{4} + \frac{1}{t} \right) + 3 \left(\frac{1}{4t^2} + \frac{1}{2t} \right)$

$$\Rightarrow \frac{dI}{dt} - \frac{3}{2} \left(\frac{t}{2} - \frac{1}{t^2} - \frac{1}{t^3} + \frac{1}{t^2} \right) - 0 \Rightarrow t^4 - 2 \Rightarrow t = \pm (2)^{\frac{1}{4}}$$

Min. $I(t) = 3 + \frac{3}{2} \left(\frac{\sqrt{2}}{4} + \frac{1}{2^{\frac{1}{4}}} + \frac{1}{2\sqrt{2}} - \frac{1}{2^{\frac{1}{4}}} \right) = 3 + \frac{3}{2} \cdot \frac{\sqrt{2}}{2} - 3 + \sqrt{\frac{9}{8}}$

$$\Rightarrow \left[\frac{9}{8} \right] + 3 = 1 + 3 = 4$$

I02/4/131. Let ' ℓ ' be tangent line at the point $T\left(t, \frac{t^2}{2}\right)$ ($t > 0$) on the curve $C: y = \frac{x^2}{2}$. A circle S touches curve C

and the x -axis. Denote by $(x(t), y(t))$ the center of circle S . If $\lim_{r \rightarrow 0^+} \int_r^1 x(t)y(t) dt = \frac{\tan \frac{3\pi}{8}}{6a}$, then the value of a is _____.

माना ' ℓ ' एक वक्र $C: y = \frac{x^2}{2}$ के किसी बिन्दु $T\left(t, \frac{t^2}{2}\right)$ ($t > 0$) पर एक स्पर्शरेखा है। एक वृत्त S वक्र C तथा x -अक्ष को

स्पर्श करता है जहाँ $(x(t), y(t))$ वृत्त S के केन्द्र के निर्देशांक हैं। यदि $\lim_{r \rightarrow 0^+} \int_r^1 x(t)y(t) dt = \frac{\tan \frac{3\pi}{8}}{6a}$ है, तब a का मान होगा।

Ans. 5

Sol. Equation of normal at point $T\left(t, \frac{t^2}{2}\right)$ to C

$$y - \frac{t^2}{t} = \frac{-1}{t}(x - t) \Rightarrow y = \frac{-x}{t} + \frac{t^2}{2} + 1$$

If circle S touches curve C at T

$(x(t), y(t))$ must lie on normal

$$\Rightarrow y(t) = \frac{-x(t)}{t} + \frac{t^2}{2} + 1$$

Also, distance of centre of circle from T equals radius = distance of centre of circle from x-axis

$$\Rightarrow (x(t) - t)^2 + \left(y(t) - \frac{t^2}{2}\right)^2 = (y(t))^2$$

$$x(t) = \frac{t(1 + \sqrt{t^2 + 1})}{2}, y(t) = \frac{t^2 + 1 - \sqrt{t^2 + 1}}{2}$$

$$\Rightarrow x(t)y(t) = \frac{1}{4}t^3\sqrt{t^2 + 1}$$

$$\lim_{t \rightarrow 0^0} \int_r^1 \frac{1}{4}t^3\sqrt{t^2 + 1}dt$$

$$\text{Let } \sqrt{1 + t^2} = z \Rightarrow 2tdt = 2zdz$$

$$\Rightarrow \lim_{t \rightarrow 0^0} \int_{\sqrt{1-t^2}}^{\sqrt{2}} \frac{1}{4}(z^2 - 1)z^2 dz = \frac{\sqrt{2} + 1}{30}$$

I02/4/132. Consider the function $f(x) = x^{\sin(x) + \cos(x)} \quad \forall x > 0$. Prove that $0.4 \leq \int_0^1 f(x)dx \leq 0.5$. _____.

फलन $f(x) = x^{\sin(x) + \cos(x)} \quad \forall x > 0$ पर विचार करें। सिद्ध कीजिए कि $0.4 \leq \int_0^1 f(x)dx \leq 0.5$. _____.

Ans.

Sol. It is easy to see that for $0 \leq x \leq 1$, we have $1 \leq \cos(x) + \sin(x) \leq \sqrt{2}$, and

$$\text{so } x^1 \geq x^{\cos(x) + \sin(x)} \geq x^{\sqrt{2}}$$

As all three functions are non-negative in $[0, 1]$, we can integrate the inequalities over that interval to get

$$\frac{1}{2} \geq \int_0^1 f(x)dx \geq \frac{1}{\sqrt{2} + 1} > \frac{1}{1.5 + 1} = 0.4$$

I02/4/133. Let $f : (0, \infty) \rightarrow \mathbb{R}$ be a continuous function such that for all $x \in (0, \infty)$, $f(2x) = f(x)$. Show that the function g defined by the equation $g(x) = \int_x^{2x} f(t) \frac{dt}{t}$ for $x > 0$ is a constant function. _____.

माना कि $f : (0, \infty) \rightarrow \mathbb{R}$ एक सतत् फलन है जो इस प्रकार है कि $f(2x) = f(x)$, $\forall x \in (0, \infty)$, तब प्रदर्शित करें कि फलन

g जो समीकरण $g(x) = \int_x^{2x} f(t) \frac{dt}{t} \quad \forall x > 0$ से निरूपित किया गया है, एक अचर फलन है।

Ans.

Sol. The function $t \rightarrow \frac{f(t)}{t}$ is continuous on $(0, \infty)$. So, g is continuously differentiable on $(0, \infty)$. We calculate the derivative of g :

$$g'(x) = \frac{d}{dx} \left(\int_x^{2x} f(t) \frac{dt}{t} \right) = 2 \frac{f(2x)}{2x} - \frac{f(x)}{x} = \frac{f(2x) - f(x)}{x} = 0.$$

So, g is a constant function.

I02/4/134. Find the value of I , where $I = \int_0^{\frac{\pi}{2}} (\cos(2\pi \sin^2 x) - 2 \cos(\pi \sin^2 x) + \cos(\pi \sin x)) dx$

I का मान ज्ञात कीजिए, जहाँ $I = \int_0^{\frac{\pi}{2}} (\cos(2\pi \sin^2 x) - 2 \cos(\pi \sin^2 x) + \cos(\pi \sin x)) dx$

Ans. 0

Sol.

I02/4/135. Evaluate the integral I , where (take $\pi = 3.14$) $I = \int_0^{\pi} e^{\cos x} \cdot \cos(\sin x) dx$

I का मान ज्ञात कीजिए, जहाँ ($\pi = 3.14$ लीजिए) $I = \int_0^{\pi} e^{\cos x} \cdot \cos(\sin x) dx$

Ans. 3.14

Sol.

I02/4/136. Let us define the integral I, as

$$I = \lim_{n \rightarrow \infty} \int_1^n \frac{x^4 + 2x^2 + 1}{\left(x^{\frac{113}{25}} + \frac{11}{3}x^{\frac{63}{25}} + 11x^{\frac{13}{25}}\right)^5} dx = \frac{p}{q}; p, q \in \mathbb{N}; \text{HCF}(p, q) = 1.$$

Number of divisors of 'p' is equal to

समाकलन I निम्न प्रकार से निरूपित किया गया है कि

$$I = \lim_{n \rightarrow \infty} \int_1^n \frac{x^4 + 2x^2 + 1}{\left(x^{\frac{113}{25}} + \frac{11}{3}x^{\frac{63}{25}} + 11x^{\frac{13}{25}}\right)^5} dx = \frac{p}{q}; p, q \in \mathbb{N}; \text{HCF}(p, q) = 1.$$

'p' के भाजकों की संख्या बराबर होगी

Ans. 10

Sol.

I02/4/137. Find the value of 3I, where I is defined as $I = \sum_{r=0}^{10} \int_r^{r+4} |x-1-r||x-2-r||x-3-r| dx$

3I का मान ज्ञात कीजिए, जहाँ I इस प्रकार से परिभाषित है कि $I = \sum_{r=0}^{10} \int_r^{r+4} |x-1-r||x-2-r||x-3-r| dx$

Ans. 165

Sol.

I02/4/138. Given that $\int_{-1}^1 \tan^{-1} \frac{1}{\sqrt{1-x^2}} \times \cot^{-1} \frac{x}{\sqrt{1-x^{2|2x|}}} dx = k\pi^2$, where $k \in \mathbb{R}$.

Then $(1-k)^2$ is equal to

दिया गया है $\int_{-1}^1 \tan^{-1} \frac{1}{\sqrt{1-x^2}} \times \cot^{-1} \frac{x}{\sqrt{1-x^{2|2x|}}} dx = k\pi^2$, जहाँ $k \in \mathbb{R}$.

तब $(1-k)^2$ का मान बराबर होगा

Ans. 0.5

Sol.

I02/4/139. Consider the following two integrals $A = \int_0^1 \sqrt{x^7(1-x)^9} dx$ and $B = \int_0^1 \sqrt{\frac{x^7(1-x)^9}{(x+5)^{20}}} dx$.

If $|\ln A - \ln B| = a \ln p_1 + b \ln p_2 + c \ln p_3$, where $p_1 < p_2 < p_3$ are prime and $a, b, c \in \mathbb{Q}$, then the value of $a - 2b + c$ is equal to

निम्न दो समाकलनों पर विचार कीजिए $A = \int_0^1 \sqrt{x^7(1-x)^9} dx$ तथा $B = \int_0^1 \sqrt{\frac{x^7(1-x)^9}{(x+5)^{20}}} dx$.

यदि $|\ln A - \ln B| = a \ln p_1 + b \ln p_2 + c \ln p_3$, जहाँ $p_1 < p_2 < p_3$ अभाज्य हैं तथा $a, b, c \in \mathbb{Q}$, तब $a - 2b + c$ का मान बराबर होगा

Ans. 1

Sol.

I03/4/140. If $y = f(x)$ is a solution of $xy^3 dy = x^2 dy + xy dx + y^4 dx$ ($x \neq 0$) where $f(1) = -2^{1/3}$ then $f^{-1}(-2)$ equals _____.

यदि $y = f(x)$ समीकरण $xy^3 dy = x^2 dy + xy dx + y^4 dx$ ($x \neq 0$) का एक हल है जहाँ $f(1) = -2^{1/3}$ है तब $f^{-1}(-2)$ बराबर होगा _____.

Ans. 4

Sol. $xy^3 dy - y^4 dx = x^2 dy + xy dx$

$$y^3 \frac{(x dy - y dx)}{x^2} = \frac{x dy + y dx}{x} \Rightarrow \left(\frac{y}{x}\right) d\frac{y}{x} = \frac{d(xy)}{(xy)^2}$$

$$\frac{1}{2} \left(\frac{y}{x}\right)^2 = \frac{-1}{xy} + C \Rightarrow \frac{1}{2} [(-2)^{1/3}]^2 = \frac{1}{2^{1/3}} + C$$

$$C = 0$$

$$\Rightarrow y^3 = -2x \Rightarrow f^{-1}(x) = -\frac{x^3}{2}$$

I03/4/141. Let $y^2(x) g(x) g'(x) + y(x) y'(x) = g(x) g'(x)$, $y(0) = 0$, $x \in \mathbb{R}$ where $f'(x)$ denotes $\frac{d}{dx} f(x)$ and $g(x)$ is a given non-constant differential function on $\mathbb{R}$ with $g(0) = g(e) = 1$. Then the value of $y(e)$ is _____.

माना $y^2(x) g(x) g'(x) + y(x) y'(x) = g(x) g'(x)$, $y(0) = 0$, $x \in \mathbb{R}$ जहाँ $f'(x)$, $\frac{d}{dx} f(x)$ को दर्शाता है तथा $g(x)$ एक अचर अवकल समीकरण है जो $\mathbb{R}$ पर परिभाषित है। यदि $g(0) = g(e) = 1$ तब $y(e)$ का मान होगा _____.

Ans. 0

Sol. Let $y^2(x) g(x) g'(x) + y(x) y'(x) = g(x) g'(x) \Rightarrow$ multiply by $2e^{g^2(x)}$

$$\Rightarrow 2y^2(x) g(x) g'(x) e^{g^2(x)} + 2y(x) y'(x) e^{g^2(x)} = 2g(x) g'(x) e^{g^2(x)}$$

$$\Rightarrow \frac{d}{dx} [e^{g^2(x)} y^2(x)] = 2g(x) g'(x) e^{g^2(x)} \Rightarrow e^{g^2(x)} y^2(x) = \int 2g(x) g'(x) e^{g^2(x)} dx$$

$$\Rightarrow g^2(x) = t \quad \Rightarrow 2g(x) g'(x) dx = dt$$

$$\Rightarrow e^{g^2(x)} y^2(x) = \int e^t dt \quad \Rightarrow e^{g^2(x)} y^2(x) = e^{g^2(x)} + c$$

$$\Rightarrow x = 0, e^{g^2(x)} y^2(0) = e^{g^2(0)} + c, g(0) = 1 \quad \Rightarrow e(0) = e + c \quad \therefore c = -e$$

Put $x = e$

$$e^{g^2(e)} y^2(e) = e^{g^2(e)} - e, g(e) = 1 \quad \Rightarrow e y^2(e) = e - e, y(e) = 0$$

I03/4/142. Let $y = f(x)$ be a curve passing through (e, e^e) which satisfy the differential equation $(2ny + xy \log_e x) dx - x \log_e x dy = 0, x > 0, y > 0$. If $g(x) = \lim_{n \rightarrow \infty} f(x)$, then $\int_{\frac{1}{e}}^e g(x) dx =$ _____.

माना कि वक्र $y = f(x)$ बिन्दु (e, e^e) से गुजरता है जो अवकल समीकरण $(2ny + xy \log_e x) dx - x \log_e x dy = 0, x > 0,$

$y > 0$ को संतुष्ट करता है। यदि $g(x) = \lim_{n \rightarrow \infty} f(x)$ है, तब $\int_{\frac{1}{e}}^e g(x) dx =$ _____.

Ans. 0

Sol. $(2ny + xy \log_e x) dx - x \log_e x dy = 0$

$$\Rightarrow \frac{dy}{y} = \left(\frac{2n}{x \log_e x} + 1 \right) dx$$

$$\Rightarrow \log y = 2n \log |\log x| + x + c \text{ \& } c = 0 \quad [\because \text{curve passes through } (e, e^e)]$$

$$\therefore y = e^{x + \log(\log x)^{2n}} = e^x (\log x)^{2n}$$

$$\Rightarrow f(x) = e^x (\log x)^{2n}$$

$$g(x) = \lim_{x \rightarrow \infty} f(x) = \begin{cases} \rightarrow \infty & \text{if } x < \frac{1}{e} \\ 0 & \text{if } \frac{1}{e} < x < e \\ \rightarrow \infty & \text{if } x > e \end{cases} \quad \therefore \int_{\frac{1}{e}}^e g(x) dx = 0$$

I03/4/143. Let $f(x)$ and $g(x)$ be two differentiable functions satisfying the following condition :

$$f(0) = 2, g(0) = 1, f(x) + g'(x) = 1 = f'(x) + g(x), x \geq 0.$$

The value of $2f(\ln 2) + 3g(\ln 3)$ is equal to

माना $f(x)$ तथा $g(x)$ दो अवकलनीय फलन हैं जो निम्न शर्त को संतुष्ट करता है :

$$f(0) = 2, g(0) = 1, f(x) + g'(x) = 1 = f'(x) + g(x), x \geq 0.$$

$2f(\ln 2) + 3g(\ln 3)$ का मान होगा

Ans. 3.5

Sol. ℓ

I01/4/144. If $f(x) = \int e^{\sin x} \left(\frac{\sin^3 x}{\sqrt{\sec^2 x + 2 \tan x \sec x}} + \frac{\sin^2 x \cos x (3 + 5 \sin x)}{(1 + 2 \sin x)^{3/2}} \right) dx$, where $f(0) = 0$, then the value of

$$\frac{256}{e} f^2 \left(\frac{\pi}{6} \right) \text{ is } \underline{\hspace{2cm}}.$$

$$\text{यदि } f(x) = \int e^{\sin x} \left(\frac{\sin^3 x}{\sqrt{\sec^2 x + 2 \tan x \sec x}} + \frac{\sin^2 x \cos x (3 + 5 \sin x)}{(1 + 2 \sin x)^{3/2}} \right) dx \text{ है, जहाँ } f(0) = 0 \text{ हो, तो } \frac{256}{e} f^2 \left(\frac{\pi}{6} \right)$$

का मान होगा _____.

Ans. 2

Sol. $\sin x = t \Rightarrow \cos x dx = dt$

$$f(x) = \int e^t \left(\frac{t^3}{\sqrt{1+2t}} + \frac{t^2(3+5t)}{(1+2t)^{3/2}} \right) dt$$

$$= e^{\sin x} \frac{\sin^3 x}{\sqrt{1+2 \sin x}} + C$$

$$f(0) = 0 \Rightarrow C = 0$$

$$f\left(\frac{\pi}{6}\right) = \frac{\sqrt{e}}{8\sqrt{2}}$$

$$\Rightarrow \frac{256}{e} f^2\left(\frac{\pi}{6}\right) = \frac{256}{e} \cdot \frac{e}{128} = 2$$

I01/4/145. If $\int \frac{\cos^2 x + \sin 2x}{(2 \cos x - \sin x)^2} dx = \frac{-A}{5} x - \frac{2}{5} \log |2 \cos x - \sin x| + \frac{1}{2 - \tan x} + C$, then A is equal to

_____.

यदि $\int \frac{\cos^2 x + \sin 2x}{(2 \cos x - \sin x)^2} dx = \frac{-A}{5} x - \frac{2}{5} \log |2 \cos x - \sin x| + \frac{1}{2 - \tan x} + C$ है तो A का मान होगा

_____.

Ans. 1

Sol. (i) Dividing the numerator and denominator by $\cos^2 x$, the given integral can be written as

$$\begin{aligned} I &= \int \frac{1 + 2 \tan x}{(2 + \tan x)^2} dx \\ &= \int \frac{\sec^2 x - \tan^2 x + 2 \tan x}{(2 - \tan x)^2} dx \\ &= \int \frac{\sec^2 x - \tan^2 x (\tan x - 2)}{(2 - \tan x)^2} dx \\ &= \int \frac{\sec^2 x}{(2 - \tan x)^2} dx + \int \frac{\tan x}{2 - \tan x} dx = I_1 + I_2 \text{ (say)} \end{aligned}$$

$$\text{Where } I_1 = \int \frac{dt}{(2 - \tan x)^2} \quad [I = \tan x]$$

$$\frac{1}{2 - t} = \frac{1}{2 - \tan x}$$

$$\text{and } I_2 = \int \frac{\tan x}{2 - \tan x} dx = \int \frac{\sin x}{2 \cos x - \sin x} dx$$

Putting $\sin x = a(2 \cos x - \sin x) + b(-2 \sin x - \cos x)$

Equation coefficients of $\sin x$ and $\cos x$, we get $0 = 2a - b$

and $1 = a - 2b$. Solving we get $a = \frac{1}{5}$ and $b = -\frac{2}{5}$

$$\therefore I_2 = -\frac{1}{5} \int \frac{2 \cos x - \sin x}{2 \cos x - \sin x} dx - \frac{2}{5} \int \frac{-2 \sin x - \cos x}{2 \cos x - \sin x} dx$$

$$= -\frac{1}{5} x - \frac{2}{5} \log |2 \cos x - \sin x| + \text{constant}$$

Thus,

$$I = \frac{1}{2 - \tan x} - \frac{1}{5} x - \frac{2}{5} \log |2 \cos x - \sin x| + C$$

I01/4/146. The value of $\int \frac{\tan x}{\tan^2 x + \tan x + 1} dx = x - \frac{2}{\sqrt{A}} \tan^{-1} \left(\frac{2 \tan x + 1}{\sqrt{A}} \right) + C$. Then the value of A is _____.

$$\int \frac{\tan x}{\tan^2 x + \tan x + 1} dx = x - \frac{2}{\sqrt{A}} \tan^{-1} \left(\frac{2 \tan x + 1}{\sqrt{A}} \right) + C \mid \text{तब } A \text{ का मान होगा } \underline{\hspace{2cm}}.$$

Ans. 3

Sol. $\int \frac{\tan x}{\tan^2 x + \tan x + 1} \cdot dx$

$$\text{let } \tan x = t$$

$$\sec^2 x dx = dt$$

$$dx = \frac{dt}{1+t^2}$$

$$\int \frac{t}{(1+t+t^2)} \cdot \frac{dt}{1+t^2} = \int \left(\frac{1}{1+t^2} - \frac{1}{1+t+t^2} \right) dt = \int \frac{dt}{1+t^2} - \int \frac{dt}{(t+1/2)^2 + \left(\frac{\sqrt{3}}{2} \right)^2}$$

$$= \tan^{-1}(t) - \frac{1}{\left(\frac{\sqrt{3}}{2} \right)} \tan^{-1} \left(\frac{2t+1}{\sqrt{3}} \right) + C = \tan^{-1}(\tan x) - \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2 \tan x + 1}{\sqrt{3}} \right) + C$$

MATCH THE COLUMN TYPE QUESTION(S)

I02/5/147. Match the following column :

Column-I

Column-II

(a) The function $f(x) = \frac{e^{x \cos x} - 1 - x}{\sin x^2}$ is not defined at $x = 0$ (p) -1

The value of $f(0)$ so that f is continuous at $x = 0$ is

(b) The value of the definite integral $\int_0^1 \frac{dx}{\sqrt{x} + \sqrt[3]{x}}$ equals $a + b \ln 2$ (q) 0

where a and b are integers then $(a + b)$ equals

(c) Given $e^n \int_0^n \frac{\sec^2 \theta - \tan \theta}{e^\theta} d\theta = 1$ then the value of $\tan(n)$ is equal to (r) $\frac{1}{2}$

(d) Let $a_n = \int_{1/n+1}^{1/n} \tan^{-1}(nx) dx$ and $b_n = \int_{1/n+1}^{1/n} \sin^{-1}(nx) dx$ then (s) 1

$\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$ has the value equal to

Choose the correct option :

निम्न स्तम्भों का सुमेलित कीजिए :

स्तम्भ-I

स्तम्भ-II

(a) फलन $f(x) = \frac{e^{x \cos x} - 1 - x}{\sin x^2}$, $x = 0$ पर परिभाषित नहीं है (p) -1

तो $f(0)$ का मान है जबकि फलन f , $x = 0$ पर सतत् है

(b) निश्चित समाकलन $\int_0^1 \frac{dx}{\sqrt{x} + \sqrt[3]{x}}$ का मान $a + b \ln 2$ है (q) 0

जहाँ a तथा b पूर्णांक हैं तो $(a + b)$ बराबर होगा

(c) दिया है $e^n \int_0^n \frac{\sec^2 \theta - \tan \theta}{e^\theta} d\theta = 1$ तो $\tan(n)$ का मान होगा (r) $1/2$

(d) यदि $a_n = \int_{1/n+1}^{1/n} \tan^{-1}(nx) dx$ और $b_n = \int_{1/n+1}^{1/n} \sin^{-1}(nx) dx$ तो (s) 1

$\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$ का मान बराबर है

सही विकल्प का चयन कीजिए :

(A) (a) - r ; (b) - p; (c) - s; (d) - r

(B) (a) - s; (b) - p; (c) - r; (d) - q

(C) (a) - p; (b) - q; (c) - r; (d) - s

(D) (a) - q; (b) - r; (c) - p; (d) - s

Ans. A

Sol. (A) $f(x) = \frac{e^{x \cos x} - 1 - x}{\sin x^2}, x \neq 0$

$f(x)$ is continuous at $x=0$

if $LHS = RHS = f(0)$

$$\begin{aligned}
 f(0) &= \lim_{x \rightarrow 0} \frac{(e^{x \cos x} - 1) - x}{\sin x^2} \\
 &= \lim_{x \rightarrow 0} \frac{\left(\left(1 + (x \cos x) + \frac{(x \cos x)^2}{2} + \dots \right) - 1 \right) - x}{\sin x^2} \\
 &= \lim_{x \rightarrow 0} \frac{x(\cos x - 1) + \frac{(x \cos x)^2}{2} + \dots}{\sin x^2} \\
 &= \lim_{x \rightarrow 0} \frac{x \left(-2 \sin^2 \frac{x}{2} \right) + \left(\frac{x \cos x}{2!} \right) + \dots}{\sin x^2} \\
 &= \lim_{x \rightarrow 0} \left(\frac{-2x \sin^2 \frac{x}{2}}{\sin x^2} + \frac{x^2 \cos^2 x}{2 \sin^2 x} \right)
 \end{aligned}$$

$$\lim_{x \rightarrow 0} \left(0 + \frac{1}{2} \right) = \frac{1}{2}$$

$$(B) \quad I = \int_0^1 \frac{dx}{\sqrt{x} + 3\sqrt{x}} = a + b \ln^2$$

$$\text{put } x = t^6 \Rightarrow dx = 6t^5 dt$$

$$I = \int_0^1 \frac{6t^5 dt}{t^3 + t^2} = 6 \int_0^1 \frac{t^3}{t+1} dt$$

$$I = 6 \int_0^1 \frac{t^3}{t+1} dt \Rightarrow I = 6 \int_0^1 \frac{t^3 + 1 - 1}{t+1} dt$$

$$= 6 \int_0^1 \left\{ (t^2 - t + 1) - \left(\frac{1}{t+1} \right) \right\} dt$$

$$= 6 \left[\frac{t^3}{3} - \frac{t^2}{2} + t - \ln^{(t+1)} \right]_0^1$$

$$= 6 \left[\frac{1}{3} - \frac{1}{2} + 1 - \ln^2 \right]$$

$$= 6 \left[\frac{2-3+6-6\ln^2}{6} \right]$$

$$I = 5 - \ln^2$$

$$a = 5, b = -6$$

$$(a + b) = -1$$

$$(c) I = e^n \int_0^n \frac{\sec^2 \theta - \tan \theta}{e^\theta} d\theta = 1$$

$$I = e^n \int_0^n e^{-\theta} (\sec^2 \theta - \tan \theta) d\theta$$

$$\text{put } \theta = -t$$

$$I = -e^n \int_0^{-n} e^t [\sec^2 t + \tan t] dt = -e^n [e^t \tan t]_0^{-n}$$

$$I = -e^n [e^{-n} \tan(-n)]$$

$$I = \tan(n)$$

$$\Rightarrow \tan(n) = 1$$

$$(D) a_n = \int_{\frac{1}{n+1}}^{\frac{1}{n}} \tan^{-1}(nx) dx, b_n = \int_{\frac{1}{n+1}}^{\frac{1}{n}} \sin^{-1}(n_x) dx$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{x \rightarrow \infty} \frac{\int_{\frac{1}{n+1}}^{\frac{1}{n}} \tan^{-1}(nx) dx}{\int_{\frac{1}{n+1}}^{\frac{1}{n}} \sin^{-1}(nx) dx}$$

$$= \lim_{n \rightarrow \infty} \left[\frac{\tan^{-1}\left(\frac{n}{n}\right) \times \left(-\frac{1}{n^2}\right) - \tan^{-1}\left(\frac{n}{n+1}\right) \times \left(\frac{-1}{(n+1)^2}\right)}{\sin^{-1}\left(\frac{n}{n}\right) \times \left(-\frac{1}{n^2}\right) - \sin^{-1}\left(\frac{n}{n+1}\right) \times \left(-\frac{1}{(n+1)^2}\right)} \right]$$

(using lebnitz rule)

$$= \lim_{n \rightarrow \infty} \left[\frac{\frac{\pi}{4} \times \frac{1}{n^2} - \tan^{-1} \left(\frac{n}{n+1} \right) \times \frac{1}{(n+1)^2}}{\frac{\pi}{2} \times \frac{1}{n^2} - \sin^{-1} \left(\frac{n}{n+1} \right) \times \frac{1}{(n+1)^2}} \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{\pi(n+1)^2 - 4n^2 \tan^{-1} \left(\frac{n}{n+1} \right)}{2 \left(\pi(n+1)^2 - 2n^2 \sin^{-1} \left(\frac{n}{n+1} \right) \right)} \right] = \frac{1}{2}$$

I02/5/148. Let $f(\theta) = \int_0^1 (x + \sin \theta)^2 dx$ and $g(\theta) = \int_0^1 (x + \cos \theta)^2 dx$ where $\theta \in [0, 2\pi]$.

The quantity $f(\theta) - g(\theta) \forall \theta$ in the interval given in column-I, is

Column-I

Column-II

(a) $\left(\frac{\pi}{4}, \frac{3\pi}{4} \right)$

(p) negative

(b) $\left(\frac{3\pi}{4}, \pi \right]$

(q) positive

(c) $\left[\frac{3\pi}{2}, \frac{7\pi}{4} \right)$

(r) non negative

(d) $\left(0, \frac{\pi}{4} \right) \cup \left(\frac{7\pi}{4}, 2\pi \right)$

(s) non positive

Choose the correct option :

माना $f(\theta) = \int_0^1 (x + \sin \theta)^2 dx$ & $g(\theta) = \int_0^1 (x + \cos \theta)^2 dx$ जहाँ $\theta \in [0, 2\pi]$. स्तम्भ -I में θ के दिये गये मानों के लिए

राशि $f(\theta) - g(\theta)$ का मान होगा। :

Column-I

Column-II

(a) $\left(\frac{\pi}{4}, \frac{3\pi}{4} \right)$

(p) ऋणात्मक

(b) $\left(\frac{3\pi}{4}, \pi \right]$

(q) धनात्मक

$$(c) \left[\frac{3\pi}{2}, \frac{7\pi}{4} \right)$$

(r) अक्रणात्मक

$$(d) \left(0, \frac{\pi}{4} \right) \cup \left(\frac{7\pi}{4}, 2\pi \right)$$

(s) अधनात्मक

सही विकल्प का चयन कीजिए :

(A) (a) – p; (b) – s; (c) – q; (d) – r

(B) (a) – s; (b) – p,q; (c) – s; (d) – r,s

(C) (a) – q,s; (b) – p,q,s; (c) – q,r; (d) – q,r

(D) (a) – q,r; (b) – r; (c) – s; (d) – p,s

Ans.

D

Sol.

$$f(\theta) = \int_0^1 (x + \sin \theta)^2 dx \text{ and } g(\theta) = \int_0^1 (x + \cos \theta)^2 dx$$

where $\theta \in [0, 2\pi]$

$$f(\theta) - g(\theta) = \int_0^1 \{x^2 + \sin^2 \theta + 2x \sin \theta\} - \{x^2 + \cos^2 \theta + 2x \cos \theta\} dx$$

$$= \int_0^1 [(\sin^2 \theta - \cos^2 \theta) + 2x(\sin \theta - \cos \theta)] dx$$

$$f(\theta) - g(\theta) = (\sin^2 \theta - \cos^2 \theta) + (\sin \theta - \cos \theta)$$

$$f(\theta) - g(\theta) = (\sin \theta - \cos \theta)[1 + \sin \theta + \cos \theta]$$

$$(A) \alpha \in \left(\frac{\pi}{4}, \frac{3\pi}{4} \right) \Rightarrow f(\theta) - g(0) \geq 0$$

Positive, non negative

$$(B) \theta \in \left(\frac{3\pi}{4}, \pi \right] \Rightarrow \sin \theta - \cos \theta > 0 \text{ and } \sin \theta + \cos \theta \in [-1, 0) \quad 1 + \sin \theta + \cos \theta \in [0, 1)$$

$$\Rightarrow f(\theta) - g(\theta) \geq 0$$

$\Rightarrow$ Non negative

$$(C) \theta \in \left[\frac{3\pi}{2}, \frac{7\pi}{4} \right)$$

$$\sin \theta + \cos \theta \in [-1, 0) \quad \text{and} \quad \sin \theta - \cos \theta < 0$$

$$1 + \sin \theta + \cos \theta \in [0, 1)$$

$$\Rightarrow f(\theta) - g(\theta) \leq 0 \quad \Rightarrow \text{Non positive}$$

$$(D) \theta \in \left(\alpha, \frac{\pi}{4} \right) \cup \left(\frac{7\pi}{4}, 2\pi \right)$$

$$\theta \in \left(0, \frac{\pi}{4} \right) \Rightarrow \sin \theta - \cos \theta < 0 \text{ and } 1 + \sin \theta + \cos \theta > 0$$

$$\Rightarrow f(\theta) - g(\theta) < 0$$

Negative, non positive

$$\theta \in \left(\frac{7\pi}{4}, 2\pi \right) \Rightarrow S + C > 0, S - C < 0$$

Negative, non positive

I02/5/149. Match the following lists :

$$I = \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{{}^n C_k}{n^k (k+r)}$$

List-I

- (a) $r = 1$
- (b) $r = 2$
- (c) $r = 3$
- (d) $r = 4$

List-II

- (p) 1
- (q) $6 - 2e$
- (r) e^{-1}
- (s) e^{-2}
- (t) e^{-3}

Choose the correct option :

निम्नलिखित सूचियों का मिलान कीजिए :

$$I = \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{{}^n C_k}{n^k (k+r)}$$

सूची-I

- (a) $r = 1$
- (b) $r = 2$
- (c) $r = 3$
- (d) $r = 4$

सूची-II

- (p) 1
- (q) $6 - 2e$
- (r) e^{-1}
- (s) e^{-2}
- (t) e^{-3}

सही विकल्प का चयन कीजिए :

- (A) a-p;b-r;c-s;d-q (B) a-r;b-p;c-s;d-q (C) a-p;b-s;c-r;d-q (D) a-p;b-s;c-q;d-r

Ans. B

Sol.
$$I = \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{{}^n C_k}{n^k (k+r)} = \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{{}^n C_k}{n^k} \int_0^1 x^{k+r-1} dx = \int_0^1 \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{{}^n C_k}{n^k} x^k \cdot x^{r-1} dx$$

$$= \int_0^1 x^{r-1} \left(\lim_{n \rightarrow \infty} \sum_{k=0}^n {}^nC_k \left(\frac{x}{n} \right)^k \right) dx = \int_0^1 x^{r-1} \left(\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n} \right)^n \right) dx$$

$$I = \int_0^1 e^x \cdot x^{r-1} dx$$

If $r = 1$; $I = \int_0^1 e^x dx = e - 1$

$$r = 2; I = \int_0^1 x e^x dx = \left[x e^x \right]_0^1 - \int_0^1 e^x dx = e - (e - 1) = 1$$

$$r = 3; I = \int_0^1 x^2 e^x dx = \left[x^2 e^x \right]_0^1 - 2 \int_0^1 x e^x dx = e - 2(1) = e - 2$$

$$r = 4; I = \int_0^1 x^3 e^x dx = \left[x^3 e^x \right]_0^1 - 3 \int_0^1 x^2 e^x dx = e - 3(e - 2) = 6 - 2e$$

I02/5/150. Match the following column :

Column-I

Column-II

(a) $\lim_{n \rightarrow \infty} \left[\prod_{r=1}^n \left(1 + \frac{r^2}{n^2} \right) \right]^{\frac{1}{n}}$

(p) $e^{\frac{3}{5}}$

(b) $\lim_{n \rightarrow \infty} \frac{(n!)^{\frac{1}{n}}}{n}$

(q) $\frac{\pi}{2}$

(c) $\lim_{n \rightarrow \infty} \sum_{r=1}^{n-1} \frac{1}{\sqrt{n^2 - r^2}}$

(r) $2e^{\frac{\pi-4}{2}}$

(d) $\lim_{n \rightarrow \infty} \left(\frac{n!}{n} \right)^{\frac{3n^2+5n}{5n^3+2n^2}}$

(s) e^{-1}

Choose the correct option :

निम्नलिखित स्तम्भों का मिलान कीजिए :

स्तम्भ-I

स्तम्भ-II

(a) $\lim_{n \rightarrow \infty} \left[\prod_{r=1}^n \left(1 + \frac{r^2}{n^2} \right) \right]^{\frac{1}{n}}$

(p) $e^{\frac{3}{5}}$

$$(b) \lim_{n \rightarrow \infty} \frac{(n!)^{\frac{1}{n}}}{n}$$

$$(q) \frac{\pi}{2}$$

$$(c) \lim_{n \rightarrow \infty} \sum_{r=1}^{n-1} \frac{1}{\sqrt{n^2 - r^2}}$$

$$(r) 2e^{\frac{\pi-4}{2}}$$

$$(d) \lim_{n \rightarrow \infty} \left(\frac{n!}{n} \right)^{\frac{3n^2+5n}{5n^3+2n^2}}$$

$$(s) e^{-1}$$

सही विकल्प का चयन कीजिए :

(A) a-s; b-r; c-q; d-p (B) a-s; b-q; c-r; d-p (C) a-r; b-s; c-q; d-p (D) a-s; b-q; c-p; d-r

Ans.

C

Sol.

$$(A) \text{ Let } y = \lim_{n \rightarrow \infty} \left[\prod_{r=1}^n \left(1 + \frac{r^2}{n^2} \right) \right]^{\frac{1}{n}}$$

$$\ell n y = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \ell n \left(1 + \frac{r^2}{n^2} \right)$$

$$\ell n y = \int_0^1 \ell n (1 + x^2) dx$$

$$\ell n y = \left[\ell n (1 + x^2) \cdot x \right]_0^1 - \int_0^1 \frac{2x^2}{1 + x^2} dx$$

$$\ell n y = \ell n 2 - 2 \int_0^1 dx + 2 \left[\tan^{-1} x \right]_0^1$$

$$\ell n y = \ell n 2 - 2 + 2 \left(\frac{\pi}{4} \right)$$

$$\ell n y = \left(\frac{\pi}{2} - 2 \right) + \ell n 2 = \ell n 2 + \ell n e^{\frac{\pi-4}{2}}$$

$$y = 2e^{\frac{\pi-4}{2}}$$

$$(B) \text{ Let } y = \lim_{n \rightarrow \infty} \left(\frac{n!}{n^n} \right)^{\frac{1}{n}}$$

$$\ell n y = \lim_{n \rightarrow \infty} \frac{1}{n} \ell n \left(\frac{1.2.3.....n}{n.n.n.....n} \right)$$

$$\ell n y = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \ell n \left(\frac{r}{n} \right)$$

$$\ell n y = \int_0^1 \ell n x dx = [x \ell n x - x]_0^1 = -1 \quad y = e^{-1}$$

$$(C) \lim_{n \rightarrow \infty} \sum_{r=1}^{n-1} \frac{1}{\sqrt{n^2 - r^2}} = \int_0^1 \frac{dx}{\sqrt{1-x^2}} = [\sin^{-1} x]_0^1 = \frac{\pi}{2}$$

$$(D) \lim_{n \rightarrow \infty} \left(\left(\frac{n!}{n^n} \right)^{\frac{1}{n}} \right)^{\frac{3n^2+5n}{5n^2+2n}} = (e^{-1})^{\frac{3}{5}} = e^{-\frac{3}{5}}$$

I02/5/151. Let $a = \int_0^1 \frac{dx}{1+(1-x)^{\frac{\pi}{2}}}$, $b = \int_0^1 \frac{dx}{1+x^4}$, $c = \int_0^1 \frac{dx}{\sqrt{4-x^2+x^5}}$, $d = \int_0^{\frac{1}{2}} \frac{dx}{\sqrt{1-x^{100}}}$

Match the truth values of entries in Column-I with those in Column-II :

Column-I

Column-II

(a) $a > \ell n 2$

(p) $c < \frac{1}{2}$

(b) $a > \frac{\pi}{4}$

(q) $c < \frac{\pi}{6}$

(c) $b < \frac{3}{4}$

(r) $d > 0.5$

(d) $b < \frac{9}{10}$

(s) $d > 0.524$

Choose the correct option :

माना $a = \int_0^1 \frac{dx}{1+(1-x)^{\frac{\pi}{2}}}$, $b = \int_0^1 \frac{dx}{1+x^4}$, $c = \int_0^1 \frac{dx}{\sqrt{4-x^2+x^5}}$, $d = \int_0^{\frac{1}{2}} \frac{dx}{\sqrt{1-x^{100}}}$

स्तम्भ-I में प्रविष्ट सही मानों का स्तम्भ-II में दी गई प्रविष्टियों से मिलान कीजिए :

स्तम्भ-I

स्तम्भ-II

(a) $a > \ell n 2$

(p) $c < \frac{1}{2}$

(b) $a > \frac{\pi}{4}$

(q) $c < \frac{\pi}{6}$

(c) $b < \frac{3}{4}$

(r) $d > 0.5$

(d) $b < \frac{9}{10}$

(s) $d > 0.524$

सही विकल्प का चयन कीजिए :

(A) a-ps,b-qr,c-ps,d-qr (B) a-ps,b-ps,c-qr,d-qr (C) a-qr,b-ps,c-ps,d-qr (D) a-ps,b-ps,c-qr,d-qr

Ans.

C

Sol.

I01/5/152. Let $f(x) = (x-1)(x-2)(x-3)\dots(x-n)$, $n \in \mathbb{N}$ and $\int \frac{f(x)f''(x) - (f'(x))^2}{f^2(x)} dx = g(x) + C$, where C

is arbitrary constant.

Column-I

Column-II

(a) If $f'(n) = 5040$, then n is divisible by

(p) 4

(b) If $g(x)$ is discontinuous at 9 points, $x \in \mathbb{R}$ then n is greater than

(q) 6

(c) If $g(x) = 5$ has atleast 8 solutions, then n may be equal to

(r) 8

(d) If the number of roots of equation $f'(x) = 0$, be $(n-5)^2(n-1)$, $(n > 1)$ (s) 9

then possible values of n is/are

Choose the correct option :

यदि $f(x) = (x-1)(x-2)(x-3)\dots(x-n)$, $n \in \mathbb{N}$ और $\int \frac{f(x)f''(x) - (f'(x))^2}{f^2(x)} dx = g(x) + C$, जहाँ C एक स्वेच्छ अचर है

स्तम्भ-I

स्तम्भ-II

(a) यदि $f'(n) = 5040$ है, तो n किस से विभाजित होगा (p)

4

(b) यदि $g(x)$, $x \in \mathbb{R}$ के 9 बिन्दुओं पर असतत् है तो n किस से बड़ा है

(q) 6

(c) यदि $g(x) = 5$ के कम से कम 8 हल है तो n बराबर हो सकता है

(r) 8

(d) यदि समीकरण $f'(x) = 0$ के हलो की संख्या $(n-5)^2(n-1)$, $(n > 1)$ है

(s) 9

तो n का/के सम्भव मान

सही विकल्प का चयन कीजिए :

(A) (a) – qs; (b) – pqs; (c) – rs; (d) – pq

(B) (a) – qs; (b) – pqs; (c) – ps; (d) – qr

(C) (a) – pr ; (b)– pqr; (c) – pq; (d) – rs

(D) (a) – pr ; (b)– pqr; (c) – rs; (d) – pq

Ans.

D

Sol. $f(x) = (x-1)(x-2)(x-3)\dots(x-n)$, $n \in \mathbb{N}$

$$\int \frac{f(x)f''(x) - (f'(x))^2}{f^2(x)} dx = g(x) + c$$

$$\int \frac{d}{dx} \left(\frac{f'(x)}{f(x)} \right) dx = g(x) + c$$

$$\frac{f'(x)}{f(x)} = g(x) + c$$

$$\ell n^{f(x)} = \ell n^{(x-1)} + \ell n^{(x-2)} + \ell n^{(x-3)} + \dots + \ell n^{(x-n)}$$

$$\frac{f'(x)}{f(x)} = \left(\frac{1}{x-1} \right) + \left(\frac{1}{x-2} \right) + \left(\frac{1}{x-3} \right) + \dots + \left(\frac{1}{x-n} \right)$$

$$f'(x) = f(x) \left[\left(\frac{1}{x-1} \right) + \left(\frac{1}{x-2} \right) + \left(\frac{1}{x-3} \right) + \dots + \left(\frac{1}{x-n} \right) \right]$$

$$(A) f'(n) = (n-1)(n-2) \dots 1 = 5040$$

$$f'(n) = \lfloor n-1 \rfloor = 5040 = \lfloor 7 \rfloor$$

$$n = 8$$

$\Rightarrow n$ is divisible by 4, 8

$$(B) g(x) = \frac{f'(x)}{f(x)} - c$$

$$g(x) = \left(\frac{1}{x-1} + \frac{1}{x-2} + \dots + \frac{1}{x-n} \right) - c$$

$g(x)$ is discontinuous at 9 points

$$\Rightarrow n \leq 9$$

$$\Rightarrow n > 4, 6, 8$$

$$(C) \text{ for } g(x) = 5$$

$$5 + c = \frac{1}{x-1} + \frac{1}{x-2} + \dots + \frac{1}{x-n}$$

$$(5 + c)(x-1)(x-2) \dots (x-n) = (x-2) \dots (n-n) + (x-1)(x-3) \dots (x-n) + \dots + (x-1)(x-n+1)$$

For at least 8 solutions degree of polynomial should be greater than equal to 8

$$(D) f'(x) = 0$$

$$\Rightarrow f(x) \left[\frac{1}{x-1} + \frac{1}{x-2} + \dots + \frac{1}{x-n} \right] = 0$$

$$\Rightarrow (x-2)(x-3)\dots(x-n) + (x-1)(x-3)\dots(x-n) + \dots + (x-1)(x-2)\dots(x-n+1) = 0$$

Number of roots = $n - 1$

$$(n-1) = (n-5)^2 (n-1)$$

$$\Rightarrow (n-1) [(n-5)^2 - 1] = 0$$

$$\therefore n \neq 1 \Rightarrow n = 4, 6$$

I01/5/153. Match the following column :

Column-I

Column-II

(a) Let $\int \cos^5 x dx = \sin x + \frac{\sin^3 x}{m} + \frac{\sin^5 x}{n} + c$, then $n + 2m$ is equal to (p) 2

(b) Let $\int \frac{xdx}{x^2 - 8x + 15} = \ell n \left| \frac{(x-5)^a}{(x-3)^b} \right| + c$, then $a + b$ is equal to (q) 3

(c) Let $y = f(t)$ & $x = g(t)$ where t is a parameter & f, g are twice derivable functions. If $f'(1) = 0$, $g'(1) = 1$ & $f''(1) = 2$ then, (r) 0

$\frac{d^2y}{dx^2}$ at $t = 1$ is equal to

(d) Let $L = \lim_{x \rightarrow 0^+} (-\ell nx)^{\frac{1}{\ell nx}}$ then $\ln L$ is equal to (s) -2

(t) Natural number

स्तम्भ-I

स्तम्भ-II

(a) माना $\int \cos^5 x dx = \sin x + \frac{\sin^3 x}{m} + \frac{\sin^5 x}{n} + c$ है, तो $n + 2m$ का मान होगा (p) 2

(b) माना $\int \frac{xdx}{x^2 - 8x + 15} = \ell n \left| \frac{(x-5)^a}{(x-3)^b} \right| + c$ हो, तो $a + b$ का मान होगा (q) 3

(c) माना $y = f(t)$ तथा $x = g(t)$ जहाँ t एक प्राचल है तथा f, g दो बार अवकलनीय फलन है यदि $f'(1) = 0$, $g'(1) = 1$ तथा $f''(1) = 2$ है, तो (r) 0

$t = 1$ पर $\frac{d^2y}{dx^2}$ का मान होगा

(d) माना $L = \lim_{x \rightarrow 0^+} (-\ell nx)^{\frac{1}{\ell nx}}$ हो, तो $\ln L$ का मान होगा (s) -2

(t) प्राकृत संख्या

(A) (a) - pt; (b) - t; (c) - pt; (d) - r

(B) (a) - pt; (b) - t; (c) - pt; (d) - t

(C) (a) - st; (b) - r; (c) - qt; (d) - qrs

(D) (a) - pt; (b) - qs; (c) - rt; (d) - rs

Ans.

A

Sol. (A) $\int \cos^5 x dx = \int (1 - \sin^2 x)^2 \cos x dx$

Let $\sin x = t$

$\Rightarrow \cos x dx = dt$

$= \int (1 - t^2)^2 dt = \int (1 + t^4 - 2t^2) dt$

$= t - \frac{2t^3}{3} + \frac{t^5}{5} + c$

$= \sin x - \frac{2}{3} \sin^3 x + \frac{\sin^5 x}{5} + c$

$\Rightarrow n + 2m = 5 - \frac{3}{2} \times 2 = 2$

(B) $I = \int \frac{x dx}{(x-3)(x-5)}$

$\Rightarrow 2I = \int \frac{(5(x-3) - 3(x-5)) dx}{(x-3)(x-5)}$

$= \ln |x-5|^5 - \ln |x-3|^3$

$I = \ln \left| \frac{(x-5)^{5/2}}{(x-3)^{3/2}} \right| + c$

$\Rightarrow a + b = 4$

(C) $\frac{dy}{dx} = \frac{f'(t)}{g'(t)}$

$\frac{d^2 y}{dx^2} = \frac{(g'(t) \cdot f''(t) - f'(t) \cdot g''(t))}{(g'(t))^2} \cdot \frac{1}{g'(t)}$

Put $t = 1$

$\frac{d^2 y}{dx^2} = \frac{2}{1} = 2$

(D) $L = \lim_{x \rightarrow 0^+} (-\ln x)^{\frac{1}{\ln x}}$

$e^{\lim_{x \rightarrow 0^+} \frac{\ln(-\ln x)}{\ln x}}$

Applying L'Hospital rule

$L = e^{\lim_{x \rightarrow 0^+} \frac{1}{-\ln x} \cdot \frac{1}{x}} = e^0 = 1$

$$\Rightarrow \ell n L = 0$$

I01/5/154. Match the following $\int f(x)dx$ is equal to, if:

Column-I

(a) $f(x) = \frac{1}{(x^2 + 1)\sqrt{x^2 + 2}}$

(b) $f(x) = \frac{1}{(x + 2)\sqrt{x^2 + 6x + 7}}$

(c) $f(x) = \frac{x^4 + x^8}{(1 - x^4)^{7/2}}$

(d) $f(x) = \sqrt{\frac{1 - \sqrt{x}}{1 + \sqrt{x}}}$

Column-II

(p) $\frac{x^5}{5(1 - x^4)^{5/2}} + C$

(q) $\sin^{-1}\left(\frac{x+1}{(x+2)\sqrt{2}}\right) + C$

(r) $(\sqrt{x} - 2)\sqrt{1 - x} + \cos^{-1}\sqrt{x} + C$

(s) $-\tan^{-1}\sqrt{1 + \frac{2}{x^2}} + C$

(t) $\frac{x^6}{6(1 - x^4)^{5/2}} + C$

निम्न को सुमेलित कीजिए

$\int f(x)dx$ बराबर है, यदि :

स्तम्भ-I

(a) $f(x) = \frac{1}{(x^2 + 1)\sqrt{x^2 + 2}}$

(b) $f(x) = \frac{1}{(x + 2)\sqrt{x^2 + 6x + 7}}$

(c) $f(x) = \frac{x^4 + x^8}{(1 - x^4)^{7/2}}$

(d) $f(x) = \sqrt{\frac{1 - \sqrt{x}}{1 + \sqrt{x}}}$

स्तम्भ-II

(p) $\frac{x^5}{5(1 - x^4)^{5/2}} + C$

(q) $\sin^{-1}\left(\frac{x+1}{(x+2)\sqrt{2}}\right) + C$

(r) $(\sqrt{x} - 2)\sqrt{1 - x} + \cos^{-1}\sqrt{x} + C$

(s) $-\tan^{-1}\sqrt{1 + \frac{2}{x^2}} + C$

(t) $\frac{x^6}{6(1 - x^4)^{5/2}} + C$

(A) $a \rightarrow p; b \rightarrow r; c \rightarrow q; d \rightarrow s$

(C) $a \rightarrow q; b \rightarrow p; c \rightarrow s; d \rightarrow t$

(B) $a \rightarrow s; b \rightarrow q; c \rightarrow p; d \rightarrow r$

(D) $a \rightarrow t; b \rightarrow s; c \rightarrow r; d \rightarrow p$

Ans. B

Sol.

$$(A) \int \frac{dx}{(x^2+1)\sqrt{x^2+2}} = \int \frac{1/x^3 dx}{\left(1+\frac{1}{x^2}\right)\sqrt{1+2/x^2}}$$

$$\text{Let } 1 + 2/x^2 = t^2$$

$$(B) I = \int \frac{1}{(x+2)\sqrt{x^2+6x+7}} dx = \int \frac{dx}{(x+2)\sqrt{(x+2)^2+2(x+2)-1}}$$

$$I = \sin^{-1} \left(\frac{x+1}{\sqrt{2(x+2)}} \right) + C$$

$$(C) \int \frac{x^4 + x^8}{(1-x^4)^{7/2}} = \int \frac{\left(x + \frac{1}{x^3}\right) dx}{\left(\frac{1}{x^2} - x^2\right)^{7/2}}$$

$$\text{Let } \frac{1}{x^2} - x^2 = t^2$$

$$(D) \text{ Let } \sqrt{x} = \cos 2\theta \Rightarrow dx = -2 \sin 4\theta d\theta$$

$$\int \sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}} dx = -2 \int \tan \theta \sin 4\theta d\theta$$

I01/5/155. Match the following column :

Column-I

(a) Let $\int \cos^5 x dx = \sin x + \frac{\sin^3 x}{m} + \frac{\sin^5 x}{n} + c$, then $n + 2m$ is equal to

(b) Let $\int \frac{xdx}{x^2-8x+15} = \ln \left| \frac{(x-5)^a}{(x-3)^b} \right| + c$, then $a + b$ is equal to

(c) Let $y = f(t)$ & $x = g(t)$ where t is a parameter & f, g are twice derivable functions. If $f'(1) = 0$, $g'(1) = 1$ & $f''(1) = 2$ then,

$$\frac{d^2y}{dx^2} \text{ at } t = 1 \text{ is equal to}$$

(d) Let $L = \lim_{x \rightarrow 0^+} (-\ln x)^{\frac{1}{\ln x}}$ then $\ln L$ is equal to

स्तम्भ-I

(a) माना $\int \cos^5 x dx = \sin x + \frac{\sin^3 x}{m} + \frac{\sin^5 x}{n} + c$, तो $n + 2m$ बराबर है

Column-II

(p) 2

(q) 3

(r) 0

(s) -2

(t) Natural number

स्तम्भ-II

(p) 2

(b) माना $\int \frac{x dx}{x^2 - 8x + 15} = \ln \left| \frac{(x-5)^a}{(x-3)^b} \right| + c$, तब $a + b$ बराबर है (q) 3

(c) माना $y = f(t)$ तथा $x = g(t)$ जहाँ t प्राचाल है तथा f, g (r) 0

दो बार अवकलनीय फलन हैं। यदि $f'(1) = 0$, $g'(1) = 1$ तथा $f''(1) = 2$, तब

$t = 1$ पर $\frac{d^2 y}{dx^2}$ बराबर है

(d) माना $L = \lim_{x \rightarrow 0^+} (-\ln x)^{\frac{1}{\ln x}}$ तो $\ln L$ बराबर है (s) -2

(t) प्राकृत संख्या

(A) (a) - (st); (b) - (t); (c) - (qt); (d) - (r) (B) (a) - (st); (b) - (t); (c) - (pt); (d) - (s)

(C) (a) - (pt); (b) - (t); (c) - (pt); (d) - (r) (D) (a) - (pt); (b) - (r); (c) - (pt); (d) - (s)

Ans. C

Sol. (A) $\int \cos^5 x dx = \int (1 - \sin^2 x)^2 \cos x dx$

Let $\sin x = t$

$\Rightarrow \cos x dx = dt$

$= \int (1 - t^2)^2 dt = \int (1 + t^4 - 2t^2) dt$

$= t - \frac{2t^3}{3} + \frac{t^5}{5} + c$

$= \sin x - \frac{2}{3} \sin^3 x + \frac{\sin^5 x}{5} + c$

$\Rightarrow n + 2m = 5 - \frac{3}{2} \times 2 = 2$

(B) $I = \int \frac{x dx}{(x-3)(x-5)}$

$\Rightarrow 2I = \int \frac{(5(x-3) - 3(x-5)) dx}{(x-3)(x-5)}$

$= \ln |x-5|^5 - \ln |x-3|^3$

$I = \ln \left| \frac{(x-5)^{5/2}}{(x-3)^{3/2}} \right| + c$

$\Rightarrow a + b = 4$

$$(C) \frac{dy}{dx} = \frac{f'(t)}{g'(t)}$$

$$\frac{d^2y}{dx^2} = \frac{(g'(t) \cdot f''(t) - f'(t) \cdot g''(t))}{(g'(t))^2} \cdot \frac{1}{g'(t)}$$

Put $t = 1$

$$\frac{d^2y}{dx^2} = \frac{2}{1} = 2$$

$$(D) L = \lim_{x \rightarrow 0^+} (-\ln x)^{\frac{1}{\ln x}}$$

$$e^{\lim_{x \rightarrow 0^+} \frac{\ln(-\ln x)}{\ln x}}$$

Applying L'Hospital rule

$$L = e^{\lim_{x \rightarrow 0^+} \frac{1}{-\ln x} \cdot \frac{1}{x} \cdot x} = e^0 = 1$$

$$\Rightarrow \ln L = 0$$

I01/5/156. Match the following column :

Column-I

Column-II

(a) Let $f(x) = \int x^{\sin x} (1 + x \cos x \log_e x + \sin x) dx$ and $f\left(\frac{\pi}{2}\right) = \frac{\pi^2}{4}$ (p) Rational

then $f(\pi) = \underline{\hspace{2cm}}$

(b) Let $g(x) = \int \frac{1 + 2 \cos x}{(\cos x + 2)^2} dx$ and $g(0) = 0$ then $g\left(\frac{\pi}{2}\right) = \underline{\hspace{2cm}}$ (q) Irrational

(c) If real numbers x and y satisfy $(x + 5)^2 + (y - 12)^2 = 14^2$ then (r) Integral
minimum value of $\sqrt{x^2 + y^2}$ is $\underline{\hspace{2cm}}$

(d) Let $k(x) = \int \frac{(x^2 + 1) dx}{\sqrt[3]{x^3 + 3x + 6}}$ and $k(-1) = \frac{1}{\sqrt[3]{2}}$, then $k(-2) = \underline{\hspace{2cm}}$ (s) Prime

निम्न को सुमेलित कीजिए :

स्तम्भ-I

स्तम्भ-II

(a) माना $f(x) = \int x^{\sin x} (1 + x \cos x \log_e x + \sin x) dx$ तथा $f\left(\frac{\pi}{2}\right) = \frac{\pi^2}{4}$ (p) परिमेय

तब $f(\pi) = \underline{\hspace{2cm}}$

(b) माना $g(x) = \int \frac{1+2\cos x}{(\cos x + 2)^2} dx$ तथा $g(0) = 0$ तब $g\left(\frac{\pi}{2}\right) = \underline{\hspace{2cm}}$ (q) अपरिमेय

(c) यदि वास्तविक संख्याएँ x तथा y , $(x+5)^2 + (y-12)^2 = 14^2$ को संतुष्ट करते हैं तब $\sqrt{x^2 + y^2}$ का न्यूनतम मान होगा $\underline{\hspace{2cm}}$ (r) पूर्णांक

(d) माना $k(x) = \int \frac{(x^2+1)dx}{\sqrt[3]{x^3+3x+6}}$ तथा $k(-1) = \frac{1}{\sqrt[3]{2}}$, तब $k(-2) = \underline{\hspace{2cm}}$ (s) अभाज्य

(A) a-p, b-q, c-p,r, d-p,r,s

(B) a-q, b-p, c-qr, d-qrs

(C) a-q, b-p, c-qr, d-qrs

(D) a-q, b-p, c-p,r, d-p,r,s

Ans.

D

Sol.

(A) $f(x) = x^{\sin x}$ and use

$$\int (f(x) + xf'(x)) dx = xf(x) + c$$

$$f\left(\frac{\pi}{2}\right) = \frac{\pi^2}{4} \Rightarrow c = 0 \Rightarrow f(\pi) = \text{irrational}$$

$$(B) \int \cos x \frac{1}{(\cos x + 2)^2} dx + \int \frac{\sin^2 x}{(\cos x + 2)^2} dx = \frac{\sin x}{2 + \cos x} + c$$

$$g(0) = 0 \Rightarrow c = 0 \quad \therefore g\left(\frac{\pi}{2}\right) = \frac{1}{2} (\text{rational})$$

$$(C) \text{ Let } \begin{cases} x+5=14\cos\theta \\ y-12=14\sin\theta \end{cases} \Rightarrow x^2 + y^2 = 365 + 28(12\sin\theta - 5\cos\theta)$$

$$\therefore \min \text{ of } \sqrt{x^2 + y^2} = \sqrt{365 - 28 \times 13} = 1$$

$$(D) t^3 = x^3 + 3x + 6$$

$$k(x) = \int t dt = \frac{t^2}{2} + c$$

$$k(-1) = \frac{1}{2}(2)^{\frac{2}{3}} + c \Rightarrow c = 0 \quad \therefore k(-2) = 2$$